Discriminative Multiple-Instance Hyperspectral Subpixel Target Characterization

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***Abstract*— Subpixel target detection in hyperspectral imagery is challenging since subpixel targets are smaller in size than the resolution of a single pixel and accurate pixel-level labels on subpixel targets are often unavailable. In particular, this article addresses the problem of learning a prime prototype target signature from imprecisely labeled highly mixed hyperspectral data. Two algorithms, multiple-instance subpixel adaptive cosine estimator (MI-SPACE) and multiple-instance subpixel spectral matched filter (MI-SPSMF), based on multiple-instance learning framework are presented. The proposed methods aim to learn a discriminative prime target signature by maximizing the posterior detection statistics of subpixel hyperspectral targets for the correspondingly proposed subpixel adaptive cosine estimator (SPACE) and subpixel spectral matched filter (SPSMF) detectors, which are also developed in this article. Experimental results demonstrate the effectiveness of the proposed methods on both simulated and real-field hyperspectral subpixel target detection tasks.**

***Index Terms*— Adaptive cosine estimator, hyperspectral, imprecise labels, multiple-instance learning, spectral matched filter, subpixel target characterization.**

I. Introduction

HYPERSPECTRAL imagery (HSI) is widely used to discover weak objects in the scene since the spectral characteristics of targets can be well represented by a number of spectral bands [1]. However, limited by low spatial resolution and compound diversity of the hyperspectral scene, each pixel is likely to be highly mixed by different ground features in HSI. Under this circumstance, subpixel target detection methods [2]–[9] are appealing. However, for most of the existing hyperspectral target detectors, certain prior knowledge of the target or even background, such as the known target signature and background space statistics, is required [10]–[14]. Common methods to obtain the desired signatures include looking up the spectral library or extracting them from a particular scene manually. Practically, the above-mentioned methods for getting spectral signatures are intractable for two reasons. First, affected by data acquisition sensor noise and environmental conditions, it often leads to the phenomenon of "same object with different spectra" between the spectral library and real HSI data. Second, pure spectral signatures extracted from the region of interest are usually desired for target detection, but one can hardly obtain a discriminative target spectrum from subpixel targets manually.

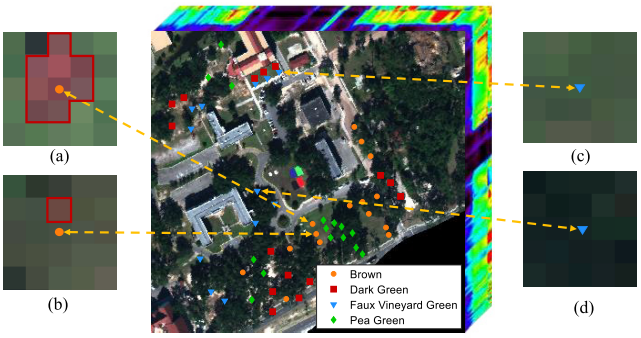


Fig. 1. Four examples of targets in the MUUFL Gulfport scene with imprecise labels for illustrating the difficulties of acquiring pixel-level labeled targets. (a) Brown target in size 3×3 m^2. (b) Brown target in size 1×1 m^2 with one-pixel ground-truth location drift. (c) and (d) Faux Vineyard Green targets with potential subpixel nature, camouflage color, and tree canopy occlusion.

In order to obtain target spectra from the region of interest adaptively, many unsupervised hyperspectral unmixing methods have been developed [15], [16]. The mixed pixels are decomposed into a collection of constituent spectra and a set of corresponding abundances. Generally, these algorithms are based on pure pixel assumption [17]–[19], minimum volume constraints [20]–[22], Bayesian approaches [23], [24], sparse regression [25], [26], or spatial–spectral contextual information [27], [28]. These unsupervised hyperspectral unmixing methods do not use the specific labeling information of a scene of interest, and thus, the endmember estimation results are generally not as discriminative as supervised learning algorithms.

However, adopting the existing supervised learning algorithms for hyperspectral target detection often requires accurately labeled targets of interest. It is often time and effort consuming to obtain precise pixel-wise labeled training targets for remotely sensed hyperspectral scene because the co-registration between the reference ground-truth information and the real location of targets could drift. Fig. 1 shows the false-color image of MUUFL Gulfport data [29] to illustrate the practical difficulties of subpixel target detection in the hyperspectral scene, collected over the University of

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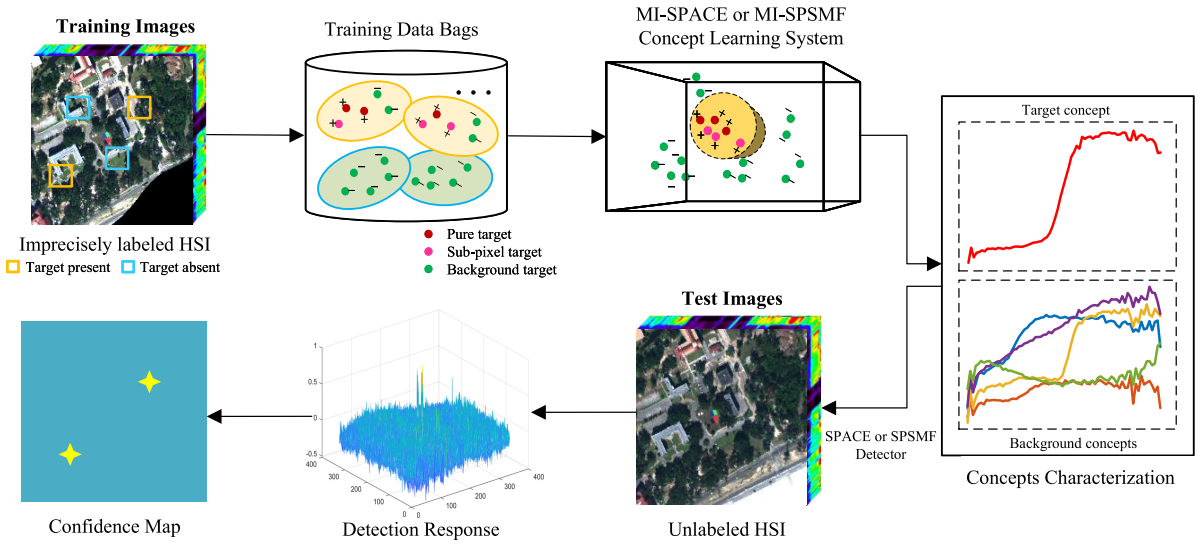


Fig. 2. Proposed multiple-instance learning subpixel target characterization framework. Hyperspectral training images with imprecise labels are divided into positive and negative bags. MI-SPACE/MI-SPSMF concept learning framework learns discriminative target and background signatures from input bags with effective background modeling. The learned signatures can be applied to the SPACE/SPSMF detector for subpixel target detection.

Southern Mississippi-Gulfpark Campus with spatial resolution 1 pixel/m^2. There are four kinds of man-made targets scattered by different colors and sizes (including several subpixel targets) throughout the scene. The four subfigures shown in Fig. 1 are four zoomed 5 $\times$ 5 target regions indicated by the global positioning system (GPS), where the orange dots or blue triangles denote the target coordinates indicated by GPS ground recording and the red rectangles denote the visually perceived presence of targets. One can visually determine that the target presence from target region (a) for this Brown target is large enough in size (3 $\times$ 3 m^2) and the denoted GPS location is accurate. Target region (b) denotes another Brown target of size 1 $\times$ 1 m^2, where one can clearly see that there exists one-pixel shift between the actual target presence and the GPS coordinate. Moreover, target regions (c) and (d) illustrate the Faux Vineyard Green targets in size 1 $\times$ 1 m^2 and 0.5 $\times$ 0.5 m^2, respectively, where it is infeasible for an analyst to discriminate pixel-level training targets under this circumstance, due to the subpixel nature, camouflage color (green targets existed in a vegetation background), and the potential tree canopy occlusion.

Therefore, the weakly supervised multiple-instance learning (MIL) [30]–[34] is desired for addressing the ambiguity existed in the labels of hyperspectral targets, which focuses on the discrimination of targets from a group of labeled areas and thus relaxes the requirement of individually (per-pixel) labeled targets. The training data are grouped into bags with knowledge of bag level labels only, e.g., positive and negative. In practice, such data bags with binary labels could be favorably obtained in two ways. One comes from the GPS coordinates (if available). The learner can take the neighborhood of the targets (region centers) with approximate GPS accuracy (region radius) as the positive bags, and the remaining parts can be divided into multiple negative bags. Besides, targets and/or backgrounds could be discriminated by visual examination or through ground reference markers by an analyst expert. The analyst could manually draw a bounding box or circle to include the region of interest containing some targets as positive bag without pointing out the exact target pixels. The regions without any salient target can be considered as negative bags. Thus, by referring to the GPS information and/or manually marking the visual targets, sufficient training samples can be obtained.

The previous contributions for addressing the above-illustrated hyperspectral detection tasks are mainly about multiple-instance hyperspectral target characterization, followed by a signature-based confidence measurement between the adaptively learned target signature and the testing pixels. Specially, the extended FUnctions of Multiple-Instances (*e*FUMI) [35] models each instance as a convex combination of positive and negative concepts and optimizes the posteriori probability of the bag-level labels using expectation–maximization. The multiple-instance hybrid estimator (MI-HE) [36] learns multiple-target and background signatures by maximizing the detection response of the hybrid subpixel detector [37], [38] under an MIL framework. In order to overcome the weakness of *e*FUMI that merges all positive bags into one and thus loses the discriminative individual bag information, the multiple-instance adaptive cosine estimator (MI-ACE) and multiple-instance spectral matched filter (MI-SMF) [39] estimate a target signature with better discriminative ability by ensuring the detection of targets from each positively labeled bags. The multiple-instance multiple diverse hyperspectral target characterization using adaptive cosine estimator (MIMD-ACE) or spectral match filter (MIMD-SMF) [40] extended the idea of discriminative target characterization to multiple-target concepts learning for better generalization ability in addressing endmember variability [41]. Similar to MI-ACE, the multiple-instance constrained energy minimization (MI-CEM) [42] learns

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a target signature by maximizing the posterior detection statistics of the constrained energy minimization subpixel detector. Book chapter [43] provides a comprehensive survey of research work on this topic.

Especially, the MI-ACE and MI-SMF [39] are representative target characterization learning methods with advantages, including discriminative ability, high training efficiency, closed-form solution, and deterministic convergence. However, it is worth noting that these methods [39], [40] assume that each of the hyperspectral data is a pure prototype of a class and does not mathematically model the subpixel property of the hyperspectral scene. Furthermore, these methods only utilized the background information (nontarget hyperspectral region) as prior knowledge, e.g., subspace whitening, failed to extract the learnable background information, and thus may suffer from subpixel hyperspectral target detection tasks with highly mixed and diverse backgrounds. Aiming at addressing these shortcomings and providing a practical solution for subpixel hyperspectral detection given only weak labels, two subpixel target characterization methods, namely, multiple-instance subpixel adaptive cosine estimator (MI-SPACE) and multiple-instance subpixel spectral matched filter (MI-SPSMF), have been developed in this article. Fig. 2 shows an illustration of the proposed model, where the convex mixing model [15] is introduced to the discriminative target characterization for subpixel target modeling and diversity promoting.

The contributions of this work are twofold. First, two subpixel detectors, subpixel adaptive cosine estimator (SPACE) and subpixel spectral matched filter (SPSMF), were developed in this article for subpixel hyperspectral target detection tasks under highly mixed and diverse backgrounds. Second, two MIL-based subpixel target characterization methods, MI-SPACE and MI-SPSMF, were proposed to address the discriminative target spectrum learning challenges from imprecisely labeled, highly mixed HSI training data with comprehensive simulated and realistic validations.

The remainder of this article is organized as follows. Section II reviews the conventional ACE and SMF detectors, as well as two MIL target characterization methods based on ACE and SMF, i.e., MI-ACE and MI-SMF. Section III first describes the proposed subpixel target detectors, SPACE and SPSMF. Subsequently, two MIL subpixel target characterization methods, MI-SPACE and MI-SPSMF, are developed based on SPACE and SPSMF frameworks. Section IV reports the experimental results and discussions, Section V investigates the robustness of our methods with respect to different parameter settings and different levels of noise, and Section VI concludes our work. The appendixes are supplementary derivations of the proposed models.

II. Related Work

This section briefly introduces the signature-based statistical detectors ACE and SMF, based on which, the framework of weakly supervised MI-ACE and MI-SMF is also derived.

A. Adaptive Cosine Estimator and Spectral Matched Filter

The adaptive cosine estimator (ACE) [10], [11], [44] and spectral match filter (SMF) [12], [13], [44], [45] are two widely applied hyperspectral target detectors, which assumes that the target and background instances follow different probability distributions and the detectors' formulation can be derived using the generalized likelihood ratio test (GLRT). Assuming the detection response, R, ACE, and SMF can be formulated as follows:

$$R_{\mathrm{ACE}}(\mathbf{s},\mathbf{x}) = \frac{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{x}}{\sqrt{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{s}}\sqrt{\mathbf{x}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{x}}} \tag{1}$$

$$R_{\mathrm{SMF}}(\mathbf{s},\mathbf{x}) = \frac{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{x}}{\sqrt{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{s}}} \tag{2}$$

where $\boldsymbol{\Sigma}_b$ is the covariance of background data, $\mathbf{s}$ is the target signature known in advance, and $\mathbf{x}$ denotes the demeaned input instance for testing. From (1) and (2), one can find that the difference between ACE and SMF is whether to normalize the input instance. Due to this difference, ACE detector evaluates the angular information between a normalized input instance and a known target signature in the whitened coordinate space, while SMF detector calculates the projection statistic of an input instance onto a known target signature in the whitened coordinate space.

B. MI-ACE and MI-SMF

Multiple-instance adaptive cosine estimator (MI-ACE) and multiple-instance spectral matched filter (MI-SMF) [39] are two efficient discriminative hyperspectral target characterization methods addressing imprecise labels based on ACE and SMF, respectively. In [39], ACE can be rewritten as follows:

$$\begin{aligned} R_{\mathrm{ACE}}(\mathbf{s},\mathbf{x}) &= \frac{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{x}}{\sqrt{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{s}}\sqrt{\mathbf{x}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{x}}} \\ &= \frac{\mathbf{s}^T\mathbf{U}\mathbf{P}^{-\frac{1}{2}}\mathbf{P}^{-\frac{1}{2}}\mathbf{U}^T\mathbf{x}}{\sqrt{\mathbf{s}^T\mathbf{U}\mathbf{P}^{-\frac{1}{2}}\mathbf{P}^{-\frac{1}{2}}\mathbf{U}^T\mathbf{s}}\sqrt{\mathbf{x}^T\mathbf{U}\mathbf{P}^{-\frac{1}{2}}\mathbf{P}^{-\frac{1}{2}}\mathbf{U}^T\mathbf{x}}} \\ &= \left(\frac{\hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|}\right)^T\left(\frac{\hat{\mathbf{x}}}{\|\hat{\mathbf{x}}\|}\right) \\ &= \hat{\hat{\mathbf{s}}}^T\hat{\hat{\mathbf{x}}} \end{aligned} \tag{3}$$

where $\mathbf{U}$ and $\mathbf{P}$ are the eigenvectors and eigenvalues of the background covariance matrix, $\boldsymbol{\Sigma}_b$. Specifically, the following substitutions are used: $\hat{\hat{\mathbf{s}}} = (\hat{\mathbf{s}}/\|\hat{\mathbf{s}}\|)$, $\hat{\mathbf{s}} = \mathbf{P}^{-(1/2)}\mathbf{U}^T\mathbf{s}$ and $\hat{\hat{\mathbf{x}}} = (\hat{\mathbf{x}}/\|\hat{\mathbf{x}}\|)$, and $\hat{\mathbf{x}} = \mathbf{P}^{-(1/2)}\mathbf{U}^T\mathbf{x}$. As shown in (3), the response of ACE is represented as the cosine similarity between the known target signature, $\mathbf{s}$, and the input test instance, $\mathbf{x}$, in the whitened coordinate space. SMF can be similarly derived as (4), where the whitened input instance $\hat{\mathbf{x}}$ is no longer normalized

$$R_{\mathrm{SMF}}(\mathbf{s},\mathbf{x}) = \hat{\hat{\mathbf{s}}}^T\hat{\mathbf{x}}. \tag{4}$$

As discussed above, it is often infeasible to obtain the target signature $\mathbf{s}$ from HSI data with imprecise ground-truth knowledge. MI-ACE and MI-SMF address this problem

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under the MIL framework, which maximizes the detection responses of targets in each positive bag $\mathbf{B}_i^+, i = 1, \ldots, K^+$ and minimizes the average detection response over all negative bags $\mathbf{B}_i^-, i = 1, \ldots, K^-$ to learn a discriminative target concept $\mathbf{s}$ without knowing the pixel-wise labels. The objective function can be expressed by

$$\mathbf{s}^* = \arg\max_{\mathbf{s}} \frac{1}{K^+}\sum_i \mathbf{P}_{\text{pos}}\left(\mathbf{s}|\mathbf{B}_i^+\right) - \frac{1}{K^-}\sum_i \mathbf{P}_{\text{neg}}\left(\mathbf{s}|\mathbf{B}_i^-\right) \quad (5)$$

where $\mathbf{P}_{\text{pos}}(\mathbf{s}|\mathbf{B}_i^+)$ and $\mathbf{P}_{\text{neg}}(\mathbf{s}|\mathbf{B}_i^-)$ are the corresponding detection statistics by applying ACE or SMF detectors over positive and negative bags, respectively. The target signature $\mathbf{s}$ can be optimized from a whitened coordinate space according to (5).

However, the subpixel detection performance of MI-ACE and MI-SMF over highly mixed data is limited by the deficiency in subpixel modeling. For example, ACE and SMF rely on the statistic background projection to dampen the response of instance falls into the background subspace. However, the response of a highly mixed weak target could also be dampened or the projection of a nontarget who is far away from the background subspace could still achieve a high target response. In the following, the subpixel ACE (SPACE) and subpixel SMF (SPSMF) have been developed with explicitly modeling of background substance mixing, based on which the MI-SPACE and MI-SPSMF have been proposed for discriminative target prototype learning from highly mixed, imprecisely labeled targets.

III. Methods

To address the deficiency of our previously proposed MI-ACE and MI-SMF for subpixel target detection tasks, the subpixel ACE (SPACE) and subpixel SMF (SPSMF) are first developed by modeling each test point as a convex combination of target and/or background concepts. SPACE and SPSMF not only inherit the advantages of ACE and SMF detectors for high efficiency, scale-invariant, but also achieve advanced discriminative ability for subpixel targets by explicitly modeling the compound background information that existed in the subpixel targets.

A. Subpixel ACE and Subpixel SMF

Assume tat each test instance, $\mathbf{x}$, can be modeled as a convex combination of a target and/or background concept set $\mathbf{D}$, where $\mathbf{D} = [\mathbf{s} \ \ \mathbf{D}^-] \in \mathbb{R}^{d\times(1+M)}$, $\mathbf{s}$ is the target concept, and $\mathbf{D}^- = [\mathbf{d}_1, \ldots, \mathbf{d}_M]$ are the M background concepts. Then, the detection problem can be posed as a binary hypothesis test with two competing hypotheses: target absent ($\boldsymbol{H}_0$) or target present ($\boldsymbol{H}_1$) shown as follows:

$$\begin{aligned} \boldsymbol{H}_1 &: \mathbf{x} = \mathbf{D}\mathbf{a} + \boldsymbol{\varepsilon} \\ &\quad = a_s\mathbf{s} + \mathbf{D}^-\mathbf{a}^- + \boldsymbol{\varepsilon} \\ \boldsymbol{H}_0 &: \mathbf{x} = \mathbf{D}^-\mathbf{a}^- + \boldsymbol{\varepsilon} \end{aligned} \quad (6)$$

where $\mathbf{a} = [a_s, \mathbf{a}^{-T}]^T$, a_s is the proportion of the target concept $\mathbf{s}$, $\mathbf{a}^- = [a_1, \ldots, a^M]^T$ is the proportion vector corresponding to the background concepts set $\mathbf{D}^-$ in $\mathbf{x}$, and $\boldsymbol{\varepsilon}$ is a random noise/error term following the zero-mean Gaussian distribution $\mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_0)$ with unknown covariance $\boldsymbol{\Sigma}_0$. The abundances $\mathbf{a}$ are subject to the sum to one and nonnegative constrains. Under this assumption, each test instance is modeled as a convex combination of the concept set $\mathbf{D}$ and the corresponding target component can be represented as (7), where $\mathbf{r}_s$ is the residual target component

$$\begin{aligned} \mathbf{r}_s &= a_s\mathbf{s} + \boldsymbol{\varepsilon} \\ &= \mathbf{x} - \mathbf{D}^-\mathbf{a}^-. \end{aligned} \quad (7)$$

Then, by substituting the test spectrum $\mathbf{x}$ in ACE and SMF by the corresponding residual target component $\mathbf{r}_s$, the proposed detectors, subpixel ACE (SPACE) and subpixel SMF (SPSMF), can be written as follows:

$$\begin{aligned} R_{\text{SPACE}}(\mathbf{s}, \mathbf{r}_s) &= \frac{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{r}_s}{\sqrt{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{s}}\sqrt{\mathbf{r}_s^T\boldsymbol{\Sigma}_b^{-1}\mathbf{r}_s}} \\ &= \frac{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\left(\mathbf{x} - \mathbf{D}^-\mathbf{a}^-\right)}{\sqrt{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{s}}\sqrt{(\mathbf{x} - \mathbf{D}^-\mathbf{a}^-)^T\boldsymbol{\Sigma}_b^{-1}(\mathbf{x} - \mathbf{D}^-\mathbf{a}^-)}} \end{aligned} \quad (8)$$

$$\begin{aligned} R_{\text{SPSMF}}(\mathbf{s}, \mathbf{r}_s) &= \frac{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{r}_s}{\sqrt{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{s}}} \\ &= \frac{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\left(\mathbf{x} - \mathbf{D}^-\mathbf{a}^-\right)}{\sqrt{\mathbf{s}^T\boldsymbol{\Sigma}_b^{-1}\mathbf{s}}}. \end{aligned} \quad (9)$$

Similarly, as described above, the formula of SPACE can be rewritten into the following form:

$$R_{\text{SPACE}}([\mathbf{s} \ \mathbf{D}^-], \mathbf{x}) = \left(\frac{\hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|}\right)^T\left(\frac{\hat{\mathbf{r}}_s}{\|\hat{\mathbf{r}}_s\|}\right) = \hat{\hat{\mathbf{s}}}^T\hat{\hat{\mathbf{r}}}_s \quad (10)$$

where

$$\hat{\hat{\mathbf{s}}} = \frac{\hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|}, \quad \hat{\mathbf{s}} = \mathbf{P}^{-\frac{1}{2}}\mathbf{U}^T\mathbf{s} \quad (11)$$

$$\hat{\hat{\mathbf{r}}}_s = \frac{\hat{\mathbf{r}}_s}{\|\hat{\mathbf{r}}_s\|}, \quad \hat{\mathbf{r}}_s = \mathbf{P}^{-\frac{1}{2}}\mathbf{U}^T\mathbf{r}_s \quad (12)$$

$\mathbf{U}$ and $\mathbf{P}$ are the eigenvectors and eigenvalues of the background covariance matrix, $\boldsymbol{\Sigma}_b$, respectively. Note that we adopt $R_{\text{SPACE}}([\mathbf{s} \ \mathbf{D}^-], \mathbf{x})$ to replace $R_{\text{SPACE}}(\mathbf{s}, \mathbf{r}_s)$ as shown in (10) in consideration of that $\mathbf{r}_s$ is a function of $\mathbf{x}$ and $\mathbf{D}^-$. Now, it can be clearly seen from the left side of the equation that the difference between ACE and SPACE is that SPACE introduces the concepts of background $\mathbf{D}^-$ for background modeling to discriminate subpixel targets.

The SPSMF detector can be similarly derived shown in (13). Appendix A provides a detailed derivation of the proposed SPACE and SPSMF detectors

$$R_{\text{SPSMF}}([\mathbf{s} \ \mathbf{D}^-], \quad \mathbf{x}) = \left(\frac{\hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|}\right)^T\hat{\mathbf{r}}_s = \hat{\hat{\mathbf{s}}}^T\hat{\mathbf{r}}_s. \quad (13)$$

The residual target component modeling of SPACE and SPSMF defined in (7) indicates that for a nontarget testing instance $\mathbf{x}$, it should be fully represented by the background concept set $\mathbf{D}^-$ and the convex code $a_s \to 0$, resulting in $\mathbf{r}_s$, falls into the background noise space with very low detector response. If $\mathbf{x}$ is a pure target prototype for test, i.e., $\mathbf{a}^- \to \mathbf{0}$,

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and thus, $\mathbf{D}^{-}\mathbf{a}^{-} \rightarrow \mathbf{0}$ and $\mathbf{r}_s \approx \mathbf{x}$, resulting in the original detector response equal to ACE and SMF. Significantly, for a subpixel target instance $\mathbf{x}$, the shape of residual target component $\mathbf{r}_s = a_s\mathbf{s} + \boldsymbol{\varepsilon}$ is approximately proportional to the known target signature $\mathbf{s}$, thus resulting in the detection responses for weak subpixel targets as salient as pure targets since SPACE is scale-invariant. Based on the above analysis, we can conclude that the detection characteristics of the SPACE and SPSMF for pure-target and nontarget test instances have been maintained and achieved advanced discriminative ability for subpixel targets.

B. MIL Subpixel Target Characterization

Let $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{d\times N}$ be training data with each instance, $\mathbf{x}_i$, being a spectral vector, where d is the number of spectral bands and N is the total number of training instances. The training data are grouped into K bags $\mathbf{B} = \{\mathbf{B}_1, \ldots, \mathbf{B}_K\}$. Each bag $\mathbf{B}_i$ containing N_i instances is associated with a binary bag-level label L_i to indicate positive or negative, where $L_i \in \{0, 1\}$ and there are K^+ positively and K^- negatively labeled bags ($K = K^+ + K^-$). An instance $\mathbf{x}_{ij} \in \mathbf{B}_i$ denotes the jth instance in bag $\mathbf{B}_i$. A positive bag (i.e., $\mathbf{B}_i$ with $L_i = 1$, denoted as $\mathbf{B}_i^+$, $i = 1, \ldots, K^+$) contains at least one instance with nontrivial target component

$$\begin{aligned} &\text{if } L_i = 1, \exists \mathbf{x}_{ij} \in \mathbf{B}_i^+ \\ &\text{s.t. } \mathbf{x}_{ij} \sim \mathcal{N}\Big(a_{sij}\mathbf{s} + \mathbf{D}^{-}\mathbf{a}_{ij}^{-}, \sigma_1^2\boldsymbol{\Sigma}_b\Big), \quad a_{sij} \neq 0. \end{aligned} \tag{14}$$

However, the number of target instances contained in one positive bag is not known. If $\mathbf{B}_i$ is negatively labeled (i.e., $L_i = 0$, denoted as $\mathbf{B}_i^-$, $i = K^+ + 1, \ldots, K$), this indicates that $\mathbf{B}_i^-$ does not contain any target

$$\text{if } L_i = 0 \forall \mathbf{x}_{ij} \in \mathbf{B}_i^- \quad \text{s.t. } \mathbf{x}_{ij} \sim \mathcal{N}\Big(\mathbf{D}^{-}\mathbf{a}_{ij}^{-}, \sigma_0^2\boldsymbol{\Sigma}_b\Big) \tag{15}$$

where σ_0 and σ_1 in (14) and (15) are two unknown noise-scaling factors, respectively.

Similar to MI-ACE and MI-SMF [39], the goal of MI-SPACE and MI-SPSMF is to learn a discriminative target signature $\mathbf{s}$ and a set of background concepts $\mathbf{D}^{-}$ that maximize the detection statistics of the corresponding SPACE and SPSMF detectors for the targets existed in positive bags and minimize the corresponding detector responses over all negative instances. This is realized by optimizing the following detection objective:

$$\begin{aligned} J_{\text{det}}([\mathbf{s}\ \mathbf{D}^{-}], \mathbf{B}) = &\frac{1}{K^+}\sum_{i=1}^{K^+} R\big([\mathbf{s}, \mathbf{D}^{-}], \mathbf{x}_i^*\big) \\ &- \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i} R([\mathbf{s}, \mathbf{D}^{-}], \mathbf{x}_{ij}) \end{aligned} \tag{16}$$

where $\mathbf{x}_i^*$ is the selected instance from each positive bag $\mathbf{B}_i^+$ that is the most prominent target prototype in the bag $\mathbf{B}_i^+$ and α is the scaling factor to balance the influence of the nontarget data to the detection objective. Given a target signature $\mathbf{s}$, the selected instance $\mathbf{x}_i^*$ is identified as the instance that achieves the highest detector response in the ith positive bag $\mathbf{B}_i^+$, shown in the following equation:

$$\mathbf{x}_i^* = \underset{\mathbf{x}_{ij}\in\mathbf{B}_i, L_i=1}{\arg\max}\ R([\mathbf{s}\ \mathbf{D}^{-}], \mathbf{x}_{ij}). \tag{17}$$

Besides the optimization of detection statistics, the minimization of reconstruction error over all negative instances term shown in (18) is included in the objective as a background data fidelity, where N^- is the total number of negative instances and $\mathbf{p}_{ij}$ is the convex codes of $\mathbf{x}_{ij}$ given $\mathbf{D}^{-}$ solved by fully constrained least-squares unmixing method (FCLS) [46], [47]. The residual target component $\mathbf{r}_s$ in the detection responses of SPACE and SPSMF depends on the compact background concepts $\mathbf{D}^{-}$. Thus, effective background estimation can further enhance the ability of the proposed detectors to suppress the background information and discover subpixel targets. With the basic assumption that the background concepts should achieve a satisfactory reconstruction of the background instances, minimizing the least-squares errors of all the negative instances is instrumental for background modeling to discriminate subpixels by utilizing $\mathbf{D}^{-}$

$$J_{\text{fid}}(\mathbf{D}^{-}, \mathbf{B}_i | L_i = 0) = \frac{1}{2N^-}\sum_{i=K^++1}^{K}\sum_{j=1}^{N_i} ||\mathbf{x}_{ij} - \mathbf{D}^{-}\mathbf{p}_{ij}||^2. \tag{18}$$

The discriminative target concept and nontarget signature set are solved by optimizing the joint objectives of the detector responses term and the fidelity term, shown as follows:

$$\begin{aligned} &\underset{\mathbf{s},\mathbf{D}^{-}}{\arg\max}\ J_{\text{det}}([\mathbf{s}\ \mathbf{D}^{-}], \mathbf{B}) - \beta J_{\text{fid}}(\mathbf{D}^{-}, \mathbf{B}_i | L_i = 0) \\ &\text{s.t. } \hat{\mathbf{s}}^T\hat{\mathbf{s}} = 1 \end{aligned} \tag{19}$$

where β is a scaling factor to balance the influence of the fidelity term on the objective. In addition, the constraint, $\hat{\mathbf{s}}^T\hat{\mathbf{s}} = 1$, resulted from the scale invariant property of SPACE and SPSMF, is applied to prevent the magnitude of $\hat{\mathbf{s}}$ from being arbitrarily large during the optimization of the detection responses term in (19).

C. Optimization

The optimization of MI-SPACE and MI-SPSMF can be decomposed into two subproblems, updating the concepts set $\mathbf{D}$ and the convex unmixing of the input data according to $\mathbf{D}$ or $\mathbf{D}^{-}$. The background concepts set $\mathbf{D}^{-}$ is initialized by using the vertex component analysis (VCA) algorithm [48] on instances over all negative bags. The desired target signature is initialized by computing the mean spectrum of instances from the positive-labeled bags.

Given (19) and the constraint $\hat{\mathbf{s}}^T\hat{\mathbf{s}} = 1$, the update equation of target signature can be derived using the Lagrange multiplier method, shown in the following:

$$\mathcal{L} = J_{\text{det}}([\mathbf{s}\ \mathbf{D}^{-}], \mathbf{B}) - \beta J_{\text{fid}}(\mathbf{D}^{-}, \mathbf{B}_i | L_i = 0) - \lambda\big(\hat{\mathbf{s}}^T\hat{\mathbf{s}} - 1\big) \tag{20}$$

where λ is the Lagrange multiplier. The update equation can be solved by taking the derivative of the Lagrangian (20) with

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respect to the target concepts $\hat{\mathbf{s}}$. For MI-SPACE, it can be derived as

$$\hat{\hat{\mathbf{s}}} = \frac{\mathbf{t}}{\|\mathbf{t}\|} \text{ where } \mathbf{t} = \frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\hat{\mathbf{r}}}_{sij} \tag{21}$$

where $\hat{\hat{\mathbf{r}}}_{si}^* = (\hat{\mathbf{r}}_{si}^*/\|\hat{\mathbf{r}}_{si}^*\|)$ and $\hat{\hat{\mathbf{r}}}_{sij} = (\hat{\mathbf{r}}_{sij}/\|\hat{\mathbf{r}}_{sij}\|)$ are the normalized residual target components under a whitened space with respect to $\hat{\mathbf{r}}_{si}^* = \mathbf{P}^{-(1/2)}\mathbf{U}^T\mathbf{r}_{si}^*$ and $\hat{\mathbf{r}}_{sij} = \mathbf{P}^{-(1/2)}\mathbf{U}^T\mathbf{r}_{sij}$, respectively. Appendix B-A provides a detailed explanation for this update process. For MI-SPSMF, the normalization of residual target component is not applied and the update equation can be written as

$$\hat{\hat{\mathbf{s}}} = \frac{\mathbf{t}}{\|\mathbf{t}\|} \text{ where } \mathbf{t} = \frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\mathbf{r}}_{sij}. \tag{22}$$

The target concept $\hat{\hat{\mathbf{s}}}$ has a closed-form solution for selected instances $\mathbf{x}_i^*$. Since there are a finite number of possibly selected instances, the number of iterations for $\hat{\hat{\mathbf{s}}}$ is also finite. In practice, the update of $\hat{\hat{\mathbf{s}}}$ usually converges in a few iterations.

The background concept set $\mathbf{D}^-$ are optimized by applying gradient ascent on the individual atom $\mathbf{d}_m$ at a time while holding the rest atoms as fixed. The gradient of the mth background concept for MI-SPACE and MI-SPSMF can be derived and shown as (23) and (24), as shown at the bottom of the page, respectively. Appendix B-B provides a derivation for solving the gradient with respect to $\mathbf{d}_m$.

Armijo's rule [49] is adopted to increase the robustness of the gradient ascent process. The convex codes $\mathbf{a}$ for all instances given $\mathbf{D}$ and the convex codes $\mathbf{p}$ for negatively labeled instance given $\mathbf{D}^-$ are solved using the FCLS unmixing method. The update iteration of MI-SPACE and MI-SPSMF stops when one of the stopping criteria is reached: 1) the difference of objective function value between two consecutive iterations δ_{iters} is less than a preset threshold θ and 2) the preset maximum iteration number $\text{iter}_{\max}$ is reached. The proposed algorithm is summarized in Algorithm 1.

Algorithm 1 MI-SPACE/MI-SPSMF

Input: Training bags and MI-SPACE/MI-SPSMF parameters
Output: $\mathbf{D}$

Initialize $\mathbf{D}^0$, $iter = 0$ and compute $\boldsymbol{\mu_b}$ and $\boldsymbol{\Sigma}_b$ as the mean and covariance of all instances in the negative bags
repeat
  Solve the corresponding convex codes $\mathbf{a}$, $\forall \mathbf{x}_{ij} \in \mathbf{B}_i, L_i = \{0, 1\}$, and $\mathbf{p}$, $\forall \mathbf{x}_{ij} \in \mathbf{B}_i^-, L_i = 0$, using FCLS unmixing method
  repeat
    Update the selected instance $\mathbf{x}_i^*$, for each positive bag according to (17)
    Update $\hat{\hat{\mathbf{s}}}$ according to (21) for MI-SPACE or (22) for MI-SPSMF
  until There is no change in the selection of $\mathbf{x}_i^*$ from each positive bag $\mathbf{B}_i^+$
  Solve $\mathbf{s}$ by inverse transformation of $\mathbf{t}$ according to $\mathbf{s} = \mathbf{U}\mathbf{P}^{\frac{1}{2}}\mathbf{t}$
  Solve the corresponding convex codes $\mathbf{a}$, $\forall \mathbf{x}_{ij} \in \mathbf{B}_i, L_i = \{0, 1\}$, using FCLS unmixing method
  for $i = 1, \ldots, M$ **do**
    Update $\mathbf{d}_m$ using gradient ascent according to (23) for MI-SPACE or (24) for MI-SPSMF
  end for
  $iter \leftarrow iter + 1$
until $(\delta_{iters} < \theta) \vee (iter \geq iter_{max})$
return $\mathbf{D}$

IV. Experiments

In this section, MI-SPACE and MI-SPSMF were evaluated and compared to several MIL-based target detection algorithms on both simulated data and real-field HSI data. The experiment on the simulation data is to gain insights on when and how the proposed algorithm is effective in the subpixel target detection tasks. The experiment on real scene HSI data is to demonstrate the practical value of the proposed algorithm. The compared MIL methods include MI-ACE and MI-SMF [39], *e*FUMI [35], EM-DD [32], MILES [34], and mi-SVM [33]. mi-SVM and MILES were included to provide comparisons with MIL classification methods that do not rely on estimating a target concept. For the experiments presented in this article, the hyperparameter settings of the proposed MI-SPACE, MI-SPSMF, and other state-of-the-art comparison algorithms were optimized by a grid search strategy. The details of the grid search and the robustness of the proposed method to varying parameters setting are presented in Section V. The receiver operating characteristic curves (ROC) with 95% confidence

$$\frac{\partial J_{\text{SPACE}}}{\partial \mathbf{d}_m} = \mathbf{U}\mathbf{P}^{-\frac{1}{2}}\left[\frac{1}{K^+}\sum_{i=1}^{K^+}\frac{a_{im}^*}{\|\hat{\mathbf{r}}_{si}^*\|}\left(\hat{\hat{\mathbf{s}}}^T\hat{\hat{\mathbf{r}}}_{si}^*\hat{\hat{\mathbf{r}}}_{si}^* - \hat{\hat{\mathbf{s}}}\right) + \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{a_{ijm}}{\|\hat{\mathbf{r}}_{sij}\|}\left(\hat{\hat{\mathbf{s}}} - \hat{\hat{\mathbf{s}}}^T\hat{\hat{\mathbf{r}}}_{sij}\hat{\hat{\mathbf{r}}}_{sij}\right)\right] + \frac{\beta}{N^-}\sum_{i=K^++1}^{K}\sum_{j=1}^{N_i}p_{ijm}(\mathbf{x}_{ij} - \mathbf{D}^-\mathbf{p}_{ij}) \tag{23}$$

$$\frac{\partial J_{\text{SPSMF}}}{\partial \mathbf{d}_m} = \mathbf{U}\mathbf{P}^{-\frac{1}{2}}\left[-\frac{1}{K^+}\sum_{i=1}^{K^+}a_{im}^*\hat{\hat{\mathbf{s}}} + \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}a_{ijm}\hat{\hat{\mathbf{s}}}\right] + \frac{\beta}{N^-}\sum_{i=K^++1}^{K}\sum_{j=1}^{N_i}p_{ijm}(\mathbf{x}_{ij} - \mathbf{D}^-\mathbf{p}_{ij}) \tag{24}$$

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TABLE I

AREA UNDER THE ROC CURVES FOR MI-SPACE, MI-SPSMF, AND COMPARISON ALGORITHMS ON RANDOMLY MIXED NOISY DATA

Algorithm	AUC	Algorithm	AUC
MI-SPSMF	**0.961**	*e*FUMI [35]	0.913
MI-SPACE	0.959	EM-DD [32]	0.873
MI-SMF [39]	0.946	mi-SVM [33]	0.927
MI-ACE [39]	0.942	MILES [34]	0.841

intervals [50] and area under the curve (AUC) were used for performance evaluation.

A. Simulated Hyperspectral Data

Simulated data were generated from four spectra selected from the ASTER spectral library [51]. In this article, the Red Slate spectrum was specified as target endmember and Verde Antique, Phyllite, and Pyroxenite were specified as background endmembers, all from rock class with 211 bands, as shown in Fig. 3. Their wavelengths are in the range of 0.4–2.5 μm.

There are two types of simulated subpixels generated: subpixels target data and subpixels nontarget data [35]. For a simulated target subpixel, m randomly selected background endmembers were used to mix with the target endmember following the linear mixing model [15]. The corresponding abundances of endmembers are sampled from the Dirichlet distribution given parameter $\boldsymbol{\alpha}_{\text{Dir}}$ as follows:

$$\boldsymbol{\alpha}_{\text{Dir}} = \begin{cases} \sigma \cdot [\alpha_{\text{mean}} \quad (1-\alpha_{\text{mean}}/m) \times \mathbf{1}_{1\times m}] \\ \qquad\qquad \text{target sub-pixel} \\ \sigma \cdot \mathbf{1}_{1\times m}, \quad \text{non-target sub-pixel} \end{cases} \tag{25}$$

where α_{mean} is the mean target proportion and σ is a parameter to control target proportion variance. The random abundances for each pixel conform the sum-to-one and nonnegative constraints. Gaussian white noise was added to control the signal-to-noise ratio (SNR) of the simulated signature. For a nontarget subpixel, only m randomly selected background endmembers and corresponding abundances sampled from the Dirichlet distribution were used to generate a mixed pixel according to the linear mixing model. Finally, the simulated mixed pixels were subgrouped into positively or negatively labeled bags according to the preset bag-level labels and the number of bag instances.

In this section, two types of simulated hyperspectral data with different levels of mixture were generated: 1) randomly mixed synthetic data that contain both pure and mixed subpixel instances and 2) highly mixed noisy data that contain only highly mixed subpixel instances.

1) Randomly Mixed Noisy Data: These data contain both randomly generated pure and mixed subpixel instances, where there are 15 positively and 5 negatively labeled bags. Each positive bag contains 100 instances among which there are 30 instances consisting of target components (i.e., true target or subpixel target) with variable proportions. For each negative bag, there are 2000 instances mixed with randomly selected background endmembers. The proportion values for mixing target and/or background endmembers are drawn from the Dirichlet distribution. The mean target proportion in each true target instance is set to 0.1. Gaussian white noise was added so that the SNR of the synthetic data is 20 dB. The training

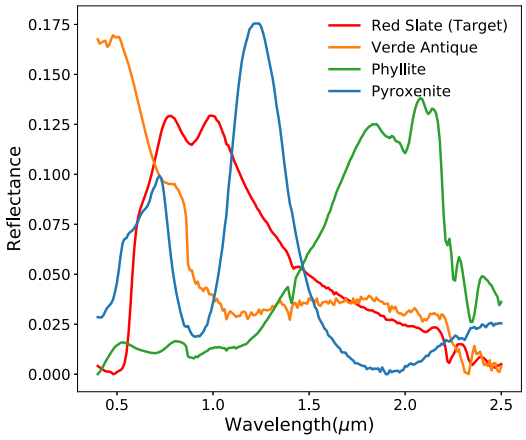


Fig. 3. Hyperspectral endmembers from ASTER spectral library used to generate hyperspectral simulated dataset.

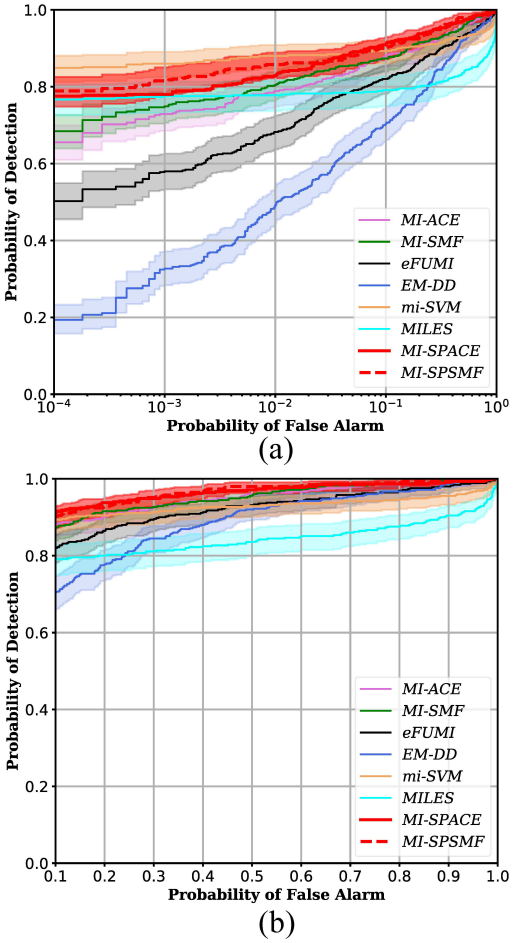


Fig. 4. ROC curves with point-wise confidence intervals of PD and FAR for MI-SPACE, MI-SPSMF, and comparison algorithms on randomly mixed noisy data. (a) Confidence intervals of PD in $\left[1\times10^{-4}, 1\right]$ FAR range. (b) Confidence intervals of PD in $\left[1\times10^{-1}, 1\right]$ FAR range. Only the best one of the proposed MI-SPACE and MI-SPSMF was plotted with confidence intervals of PD due to the limitation of drawing space.

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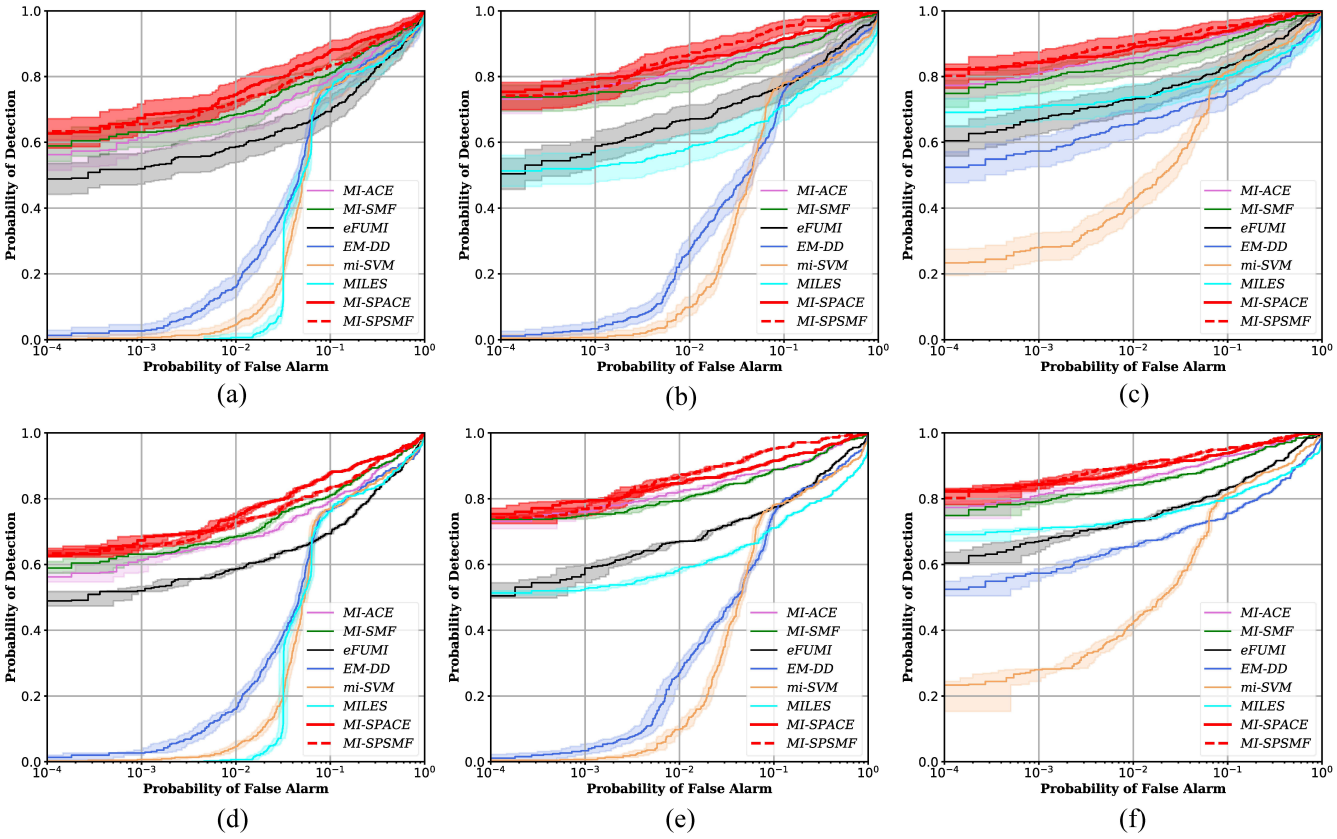


Fig. 5. ROC curves with point-wise confidence intervals of PD and FAR for MI-SPACE, MI-SPSMF, and comparison algorithms on highly mixed noisy datasets with varied mean target proportion α_{mean}. (a) Confidence intervals of PD and $\alpha_{\text{mean}} = 0.1$. (b) Confidence intervals of PD and $\alpha_{\text{mean}} = 0.2$. (c) Confidence intervals of PD and $\alpha_{\text{mean}} = 0.3$. (d) Confidence intervals of FAR and $\alpha_{\text{mean}} = 0.1$. (e) Confidence intervals of FAR and $\alpha_{\text{mean}} = 0.2$. (f) Confidence intervals of FAR and $\alpha_{\text{mean}} = 0.3$. Only the best one of the proposed MI-SPACE and MI-SPSMF was plotted with confidence intervals of PD due to the limitation of drawing space.

data and testing data are generated separately using the same generating procedure. During the testing process, SPACE was applied as the detector for signatures learned from MI-SPACE and MI-SPSMF. In contrast, the ACE detector is used for signatures estimated from MI-ACE, MI-SMF, *e*FUMI, and EM-DD as a comparison.

The optimized parameter settings of MI-SPACE and MI-SPSMF for this experiment are $\alpha = 0.5$, $\beta = 1 \times 10^3$, and $M = 3$. The performance was evaluated using the ROC curves with confidence intervals of PD and AUC reported in Fig. 4(a) and Table I. The results shown for each method are the median value over five runs. It can be found in Fig. 4(a) that even though our proposed algorithm has the best overall performance according to the AUC values, mi-SVM is better in low FAR range, especially in $[0, 10^{-2}]$ FAR region. This is due to the configuration of this random mixed simulated experiment containing pure target pixels in the positive bags. mi-SVM performs very well on these pure pixel targets in the beginning of FAR but has lower performance on the remaining subpixel targets compared with the proposed methods, as reflected in the overall AUC values (mi-SVM: 0.927, proposed MI-SPSMF: 0.961, and MI-SPACE: 0.959). More specifically, we carefully examined and plotted the ROC curves with point-wise confidence intervals of PD in $[10^{-1}, 1]$ FAR range as shown in Fig. 4(b), which reflects the overall leading performance of the proposed algorithms on subpixel targets compared with other MIL based methods.

TABLE II

AREA UNDER THE ROC CURVES (AUC) FOR SPSMF, SPACE, AND COMPARISON DETECTORS USING GROUND TRUTH ON RANDOMLY MIXED NOISY DATA. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	AUC	Algorithm	AUC
SPSMF+Ground Truth	**0.991**	SMF+Ground Truth [12]	0.986
SPACE+Ground Truth	<u>0.989</u>	ACE+Ground Truth [10]	0.987

For simulated experiments on randomly mixed noisy datasets, we also conducted a comparison between the proposed subpixel detectors (SPACE and SPSMF) and the classic prime signature-based detectors (ACE and SMF) with ground-truth signatures, as reported in Table II. As can be clearly seen from Table II, the proposed detectors achieved the overall better performance than ACE and SMF under these randomly mixed target detection configurations.

2) Highly Mixed Noisy Data: This experimental configuration provides the validation for the proposed method on highly mixed data, where each target instance was mixed with at least one background component, resulting in a challenging noisy and subpixel target detection task. The mean target proportion

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TABLE III
LIST OF CONSTITUENT ENDMEMBERS FOR HIGHLY MIXED NOISY DATA

Bag Index	Bag Label	Instance Num.(Target Num.)	Target Endmember	Background Endmember
1-5	+	100 (30)	Red Slate	Verde Antique, Phyllite, Pyroxenite
6-10	+	100 (30)	Red Slate	Phyllite, Pyroxenite
11-15	+	100 (30)	Red Slate	Pyroxenite
16-20	−	2000 (0)	N/A	Verde Antique, Phyllite, Pyroxenite

value α_{mean} for varying the level of subpixel target component from weak to high was set to 0.1, 0.2, 0.3, and 0.5. The level of background mixture was also varied to formulate a more difficult detection task with incomplete background knowledge in the positive bags. The formulation of highly mixed subpixel detection tasks was listed in Table III. The other settings were identical to the previous randomly mixed detection task.

The parameter settings of MI-SPACE and MI-SPSMF for this experiment are $\alpha = 0.5$, $\beta = 1 \times 10^3$, and $M = 3$. Fig. 5 presents the detection ROC curves with confidence intervals of PD and FAR. Table IV records the quantitative AUC values of the proposed MI-SPACE, MI-SPSMF, and comparison algorithms. The reported results are the median value over five runs. As shown in Fig. 5(d)–(f), confidence intervals of FAR for both MI-SPACE and MI-SPSMF are distinctly separated from other comparison algorithms, which proves a statistically significant difference between the proposed methods and comparison approaches. From the perspective of confidence intervals of PD shown in Fig. 5(a)–(c), it can be similarly found that confidence intervals by the proposed MI-SPACE and MI-SPSMF have little overlap with the other comparison algorithms and the ROC curves are also evidently higher, which validates the effectiveness of the proposed methods from the perspective of confidence interval.

The average execution time for MI-SPSMF, MI-SPACE, and comparison algorithms over five runs is listed in the right column of Table IV. It can be observed that all the tested methods are time-efficient. MI-SPACE and MI-SPSMF methods are more time-consuming than MI-ACE and MI-SMF for the proposed methods require more iterations to learn multiple background concepts, which facilitates background modeling and subpixel targets discrimination. They are still more efficient than the other two multiple-instance concept learning methods, *e*FUMI and EM-DD.

The initialization of target concept $\mathbf{s}$ in D is conducted by computing the mean spectrum of the union of all positively labeled training bags. Fig. 6 shows the initial target signature (dotted red) averaged from the positive bags of highly mixed noisy data with 0.2 mean target proportion, where the nontarget and subpixel target pixels are plotted in gray and blue, respectively. One can clearly see that the initial spectrum is flat and drifted by the dominant number of nontarget pixels mixed in positive bags. As a comparison, Fig. 7 plots together the initial target spectrum and the learned target and background signatures by MI-SPACE, where one can clearly find that the target signature learned by the proposed method captures the distinct target response in wavelength range 0.7–1.2 μm, validating the discriminative ability of the proposed methods.

TABLE IV
AREA UNDER THE ROC CURVES (AUC) AND AVERAGE EXECUTION TIME OVER FIVE RUNS FOR MI-SPSMF, MI-SPACE, AND COMPARISON ALGORITHMS ON HIGHLY MIXED NOISY DATASET WITH VARIED MEAN TARGET PROPORTIONS. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	α_{mean}				Avg. Time (s)
	0.1	0.2	0.3	0.5	
MI-SPSMF	<u>0.922</u>	**0.979**	**0.982**	**0.999**	17.467
MI-SPACE	**0.926**	<u>0.962</u>	<u>0.978</u>	**0.999**	25.937
MI-SMF [39]	0.909	0.949	0.964	<u>0.997</u>	9.273
MI-ACE [39]	0.905	0.951	0.968	0.990	9.296
*e*FUMI [35]	0.869	0.893	0.927	0.980	37.893
EM-DD [32]	0.851	0.856	0.869	0.991	41.993
mi-SVM [33]	0.842	0.849	0.895	0.947	19.432
MILES [34]	0.813	0.823	0.880	0.967	6.944

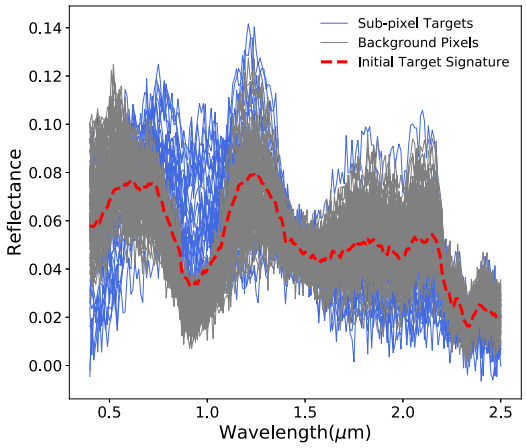


Fig. 6. Initial target signature (dotted red) computed from the mean spectrum of all positive training bag (gray) of highly mixed noisy datasets with 0.2 mean target proportion.

We also conducted a comparison between the proposed subpixel detectors (SPACE and SPSMF) and the classic prime signature-based detectors (ACE and SMF) on the highly mixed noisy data with varied mean target proportion $\alpha_{\text{mean}} = 0.1$, 0.2, 0.3, and 0.5. Ground truth signatures were applied to these detectors for target detection and the performance results are reported in Table V. As can be observed from Table V, given the known ground-truth spectra, the proposed detectors achieved the overall better performance than ACE and SMF under these highly mixed target detection configurations.

B. MUUFL Gulfport Hyperspectral Data

A real-field HSI target detection task, MUUFL Gulfport hyperspectral target detection campaign [29], has been included to evaluate the proposed methods. This dataset was

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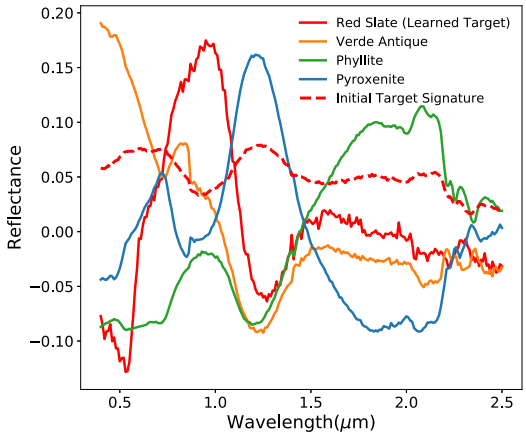


Fig. 7. Learned discriminative target and background concepts by MI-SPACE from highly mixed noisy datasets with 0.2 mean target proportion.

TABLE V

AREA UNDER THE ROC CURVES (AUC) FOR SPSMF, SPACE, AND COMPARISON DETECTORS USING GROUND TRUTH ON HIGHLY MIXED NOISY DATASET WITH VARIED MEAN TARGET PROPORTIONS. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	α_{mean}			
	0.1	0.2	0.3	0.5
SPSMF+Ground Truth	<u>0.942</u>	<u>0.969</u>	**0.996**	<u>0.999</u>
SPACE+Ground Truth	**0.951**	**0.978**	<u>0.995</u>	**1**
SMF+Ground Truth [12]	0.920	0.958	0.987	<u>0.999</u>
ACE+Ground Truth [10]	0.925	0.957	0.986	<u>0.999</u>

TABLE VI

INFORMATION OF EMLACED TARGETS IN THE MUUFL GULFPORT DATASET

Target Type	Num. of different size targets			Total Num.
	$0.5m\times0.5m$	$1m\times1m$	$3m\times3m$	
Brown	5	5	5	15
Dark Green	5	5	5	15
Faux Vineyard Green	4	4	4	12
Pea Green	5	5	5	15

collected over the campus of the University of Southern Mississippi at different heading degrees on November 8, 2010. Five flights were captured. The range of the hyperspectral scene is approximately 330 m $\times$ 340 m with spatial resolution 1 pixel/m^2. Each pixel is with 72 spectral bands corresponding to wavelength 375–1050 nm at a 10-nm sampling interval. Four types of man-made targets were emplaced around the scene and the ground-truth locations were recorded by a GPS device. These targets are square cloth plastic tarps of four different colors: Brown (15 examples), Dark Green (DG) (15 examples), Faux Vineyard Green (FVG) (12 examples), and Pea Green (PG) (15 examples) with varied sizes from 0.25 m^2, 1 m^2, and 9 m^2 in area, as shown in Table VI and Fig. 8. In addition, there existed heavy trees coverage around the scene and some of the targets were occluded by live Oak trees, resulting in a challenging hyperspectral subpixel target detection task.

1) MUULF Gulfport Mixed Targets Validation: In this experimental configuration, two images, Flight 1 and Flight 3, collected over the same scene and altitude at different times were included as cross-validated data for training and testing. Each target presence indicated by GPS was treated as a positive bag where the 5 $\times$ 5 region centered around the GPS

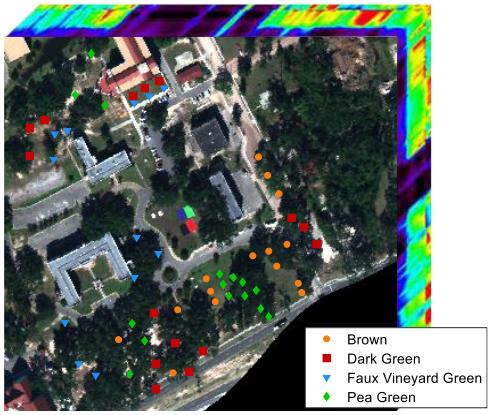


Fig. 8. MUUFL Gulfport hyperspectral false-color image and 57 target locations of four categories.

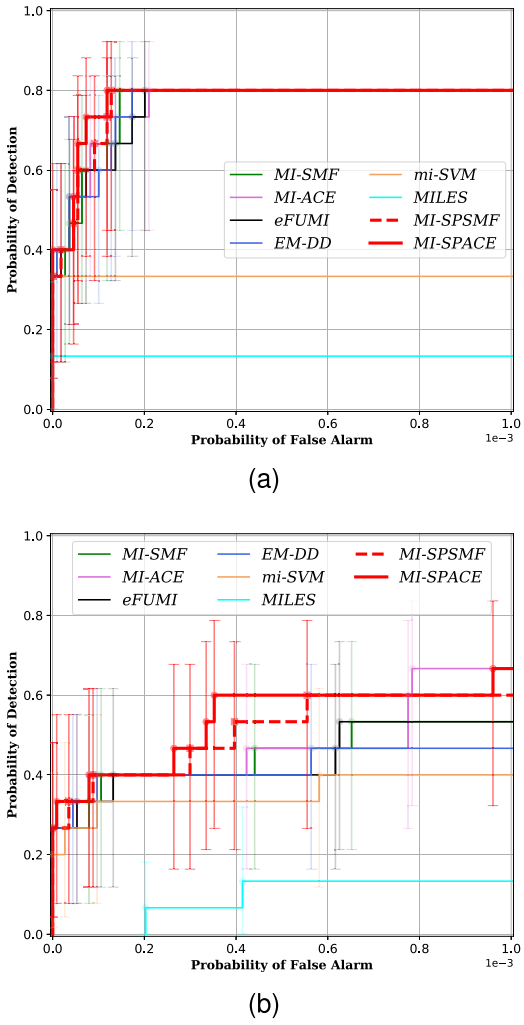


Fig. 9. ROC curves with point-wise confidence intervals of PD for MI-SPSMF, MI-SPACE, and comparisons on MUUFL Gulfport dataset at FAR = 1×10^{-3}, brown-type targets. (a) Training on flight 3, testing on flight 1. (b) Training on flight 1, testing on flight 3.

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TABLE VII

NORMALIZED AREA UNDER THE ROC CURVES (NAUC) AT FAR $= 1 \times 10^{-3}$ FOR MI-SPSMF, MI-SPACE, AND COMPARISON ALGORITHMS ON THE MUUFL GULFPORT DATA

Algorithm	Train on Flight 1; Test on Flight 3				Train on Flight 3; Test on Flight 1			
	Brown	DG	FVG	PG	Brown	DG	FVG	PG
MI-SPSMF	<u>0.508</u>	**0.401**	<u>0.633</u>	0.331	<u>0.766</u>	<u>0.501</u>	**0.638**	0.388
MI-SPACE	**0.533**	<u>0.396</u>	0.615	**0.344**	**0.774**	**0.512**	<u>0.621</u>	<u>0.401</u>
MI-SMF [39]	0.448	0.382	0.579	0.316	0.760	<u>0.501</u>	0.613	0.388
MI-ACE [39]	0.474	0.390	0.485	<u>0.333</u>	0.760	0.483	0.593	0.380
eFUMI [35]	0.423	0.377	**0.654**	0.267	0.754	0.491	0.605	0.393
EM-DD [32]	0.411	0.381	0.486	0.279	0.760	0.503	0.541	**0.416**
mi-SVM [33]	0.353	0.195	0.412	0.265	0.333	0.319	0.245	0.274
MILES [34]	0.092	0.193	0.219	0.091	0.133	0.294	0.123	0.080

TABLE VIII

NAUC AT FAR $= 1 \times 10^{-3}$ FOR SPSMF, SPACE, AND COMPARISON DETECTORS ON MUUFL GULFPORT DATA. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	Test on Flight 3				Test on Flight 1			
	Brown	DG	FVG	PG	Brown	DG	FVG	PG
SPSMF+Ground Truth+VCA	<u>0.424</u>	0.362	**0.464**	**0.301**	0.677	0.474	0.390	0.355
SPACE+Ground Truth+VCA	**0.438**	**0.396**	<u>0.455</u>	<u>0.294</u>	**0.763**	**0.519**	**0.535**	**0.414**
SMF+Ground Truth [12]	0.375	0.354	0.406	0.264	0.643	0.433	0.356	0.381
ACE+Ground Truth [10]	0.421	<u>0.380</u>	0.409	0.267	<u>0.751</u>	<u>0.513</u>	<u>0.426</u>	<u>0.392</u>

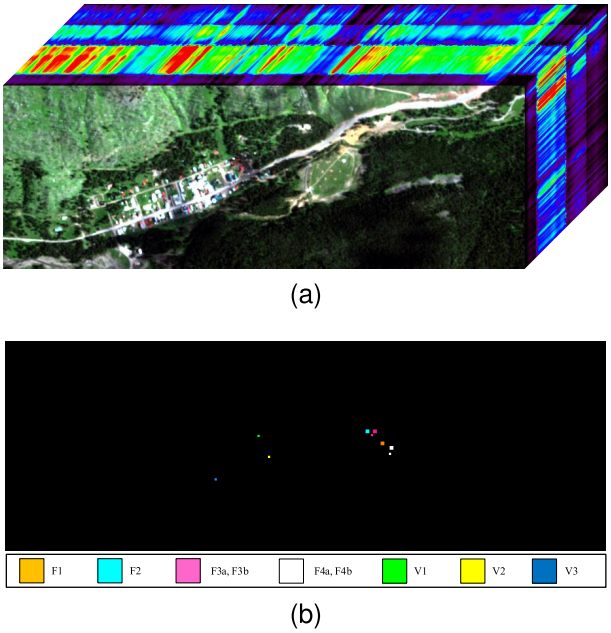


Fig. 10. HyMap dataset scenario display. (a) False color rendering of self-test hyperspectral image in the HyMap dataset, with size of 280 × 800 × 126. (b) Ground truth locations of nine targets marked with different colors and different sizes.

coordinate was grouped as a positive bag, as shown in Fig. 2. The rest image area that did not have any type of target was grouped as a big negative bag. In this way, the target detection from approximate GPS ground truth has been modeled as an MIL problem according to the GPS spatial deviation.

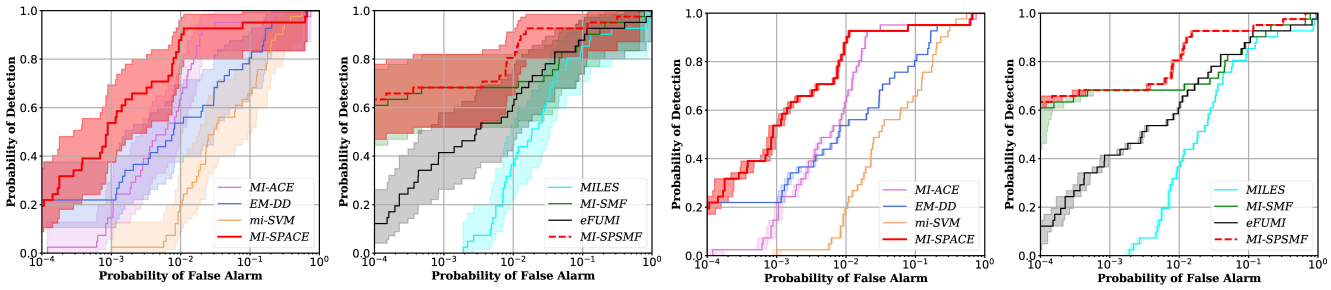
The parameter settings of MI-SPACE and MI-SPSMF for this experiment are $\alpha = 0.3$, $\beta = 1 \times 10^6$, and $M = 10$. The normalized area under the receiver operating characteristic curve (NAUC) was adopted as the performance evaluation metric in which the area was normalized out to a false alarm rate (FAR) at 1×10^{-3} [52], in order to emphasize the sensitivity of performance to false alarms.

Table VII summarizes the quantitative evaluation (NAUC) on this real-field detection task cross-validated between Flight 1 and 3. The results reported are the median value over five runs. The ROC curves with confidence intervals of PD at FAR $= 1 \times 10^{-3}$ for detected Brown target pixels are shown in Fig. 9, where one can see that the proposed methods outperform mi-SVM and EM-DD significantly. For the other comparisons that show no statistically significant difference from the confidence interval perspective, the proposed methods still achieved a better NAUC performance.

For MUULF Gulfport mixed targets validation, we also conducted a comparison of detectors. During this experiment, four ground-truth target spectra manually selected from each of Flights 1 and 3 were given as prior knowledge for cross validation. Background spectra learned by unsupervised method VCA [48] have also been acquired to form a fair comparison. The detection statistics were recorded in Table VIII. It can be observed that the proposed SPACE detector obviously achieved better performance than the other comparisons. In addition, we noticed that the proposed SPSMF and SPACE detectors perform significantly better over the traditional detectors given (weak) manually selected target spectra for their better discriminative and background modeling abilities.

2) MUULF Gulfport Subpixel Validation: In this experimental setting, we combined the subpixel targets across the five flight scenes in MUULF Gulfport Data together to construct a subpixel target detection task. Specifically, for each type of targets among the five flights, only the targets with size smaller or equal to 1 m × 1 m are retained according to the ground-truth locations. Here, we treat the targets of size less or equal to 1 m × 1 m as subpixels since the spatial resolution of the image is 1 pixel/m^2 and the targets are

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(a) (b) (c) (d)

Fig. 11. ROC curves with point-wise confidence intervals of PD and FAR on the HyMap dataset. (a) Confidence intervals of PD for MI-ACE, EM-DD, mi-SVM, and proposed MI-SPACE. (b) Confidence intervals of PD for MI-SMF, *e*FUMI, MILES, and proposed MI-SPSMF. (c) Confidence intervals of FAR for MI-ACE, EM-DD, mi-SVM, and proposed MI-SPACE. (d) Confidence intervals of FAR for MI-SMF, *e*FUMI, MILES, and proposed MI-SPSMF.

TABLE IX

NAUC AT FAR $= 1 \times 10^{-3}$ FOR MI-SPSMF, MI-SPACE, AND COMPARISON ALGORITHMS ON MUUFL GULFPORT SUBPIXEL TARGETS DETECTION. TRAINING ON FLIGHTS 1–3 AND TESTING ON FLIGHTS 4 AND 5. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	Brown	DG	FVG	PG
MI-SPSMF	**0.354**	**0.266**	0.144	0.170
MI-SPACE	0.333	0.259	**0.151**	**0.210**
MI-SMF [39]	0.308	0.231	0.117	0.076
MI-ACE [39]	0.310	0.252	0.118	0.158
*e*FUMI [35]	0.182	0.199	0.016	0.072
EM-DD [32]	0.234	0.069	0.055	0.024
mi-SVM [33]	0.036	0.098	0.117	0.042
MILES [34]	0.019	0.215	0.116	0.039

rarely aligned with the pixel grid, resulting in these target responses taking across several pixels and being subpixels. The information of targets in each image can be found in Table VI. Similarly, each target presence indicated by GPS was treated as a positive bag where the 5×5 region centered around the GPS coordinate was grouped together resulting in 38 subpixel targets of four different colors for each image: Brown (ten examples), Dark Green (DG) (ten examples), Faux Vineyard Green (FVG) (eight examples), and Pea Green (PG) (ten examples). Flights 1–3 were used for training, and Flights 4 and 5 were used for testing. The parameter settings of MI-SPACE and MI-SPSMF for this experiment are $\alpha = 0.6$, $\beta = 1 \times 10^5$, and $M = 8$. NAUC at FAR $= 1 \times 10^{-3}$ is adopted as the performance evaluation metrics. The experiment results are shown in Table IX, where the best results were shown in bold and the second best results were underlined. As can be seen from Table IX, the proposed MI-SAPCE and MI-SPSMF outperform MI-ACE and other comparisons significantly in this subpixel task due to the target concept learning and background modeling abilities for subpixel targets.

We again included a comparison between the proposed subpixel detectors, SPACE and SPSMF, and the classic detectors, ACE and SMF, in this subpixel target detection task. Since these signature-based detectors require the prior target and/or background knowledge (spectra), the ground-truth target spectra provided by the MUUFL Gulfport data and the background spectra learned by the unsupervised method VCA [48] have been acquired to form a fair comparison. Table X shows

TABLE X

NAUC AT FAR $= 1 \times 10^{-3}$ FOR SPSMF, SPACE, AND COMPARISON DETECTORS ON MUUFL GULFPORT SUBPIXEL TARGET DETECTION. TESTING ON FLIGHTS 4 AND 5. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	Brown	DG	FVG	PG
SPSMF+Ground Truth+VCA	0.184	0.115	0.078	0.102
SPACE+Ground Truth+VCA	**0.288**	**0.234**	**0.142**	**0.124**
SMF+Ground Truth [12]	0.119	0.043	0.024	0.062
ACE+Ground Truth [10]	0.179	0.140	0.024	0.053

the performance of the proposed and comparison detectors using the off-the-shelf spectra on this real-field subpixel target detection task, where one can clearly see that the proposed subpixel detectors outperform the existing ACE and SMF detectors, proving its superiority for subpixel target detection scenario.

3) Analysis of Bag Generation Without GPS Information: In some real scenarios, the hyperspectral images may lack of GPS coordinates and even most targets of interest are small in size. It leads to a dilemma that an analyst can only determine the locations of few targets in training data. It is meaningful to study a new approach to generating MIL data bags without GPS information.

For the real-field MUULF Gulfport dataset, three spectra for each type of target manually selected from the scene are given as prior knowledge. Then, we use the ACE detector [10] with those ground-truth target spectra for preprocessing to get three preliminary detection maps for each type of target in training data. The confidence of each pixel can be obtained by a point-wise maximizing operation on those three maps. Since there are only 12 or 15 weak target presences of each target type, the top 100 points with higher confidence are considered to include most targets of interest and treated as weak knowledge of the target presence. Multiple 5×5 bounding boxes centered by these 100 salient points were generated as positive bags. The overlapped bounding boxes were grouped together, which means that the number of generated positive bags could be much less. The rest image area was grouped as a big negative bag.

The parameter settings of MI-SPACE and MI-SPSMF for this experiment are $\alpha = 0.3$, $\beta = 1 \times 10^6$, and $M = 10$. Flights 1 and 3 were included as cross-validated data for training and

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TABLE XI

NORMALIZED AREA UNDER THE ROC CURVES (NAUC) AT FAR $= 1\times10^{-3}$ FOR MI-SPSMF, MI-SPACE, AND COMPARISON ALGORITHMS ON MUUFL GULFPORT DATA. NO GPS INFORMATION DURING THE TRAINING PROCESS

Algorithm	Train on Flight 1; Test on Flight 3				Train on Flight 3; Test on Flight 1			
	Brown	DG	FVG	PG	Brown	DG	FVG	PG
MI-SPSMF	0.436	0.374	0.637	**0.267**	0.732	0.484	**0.598**	**0.397**
MI-SPACE	**0.467**	0.381	**0.640**	**0.267**	**0.737**	**0.486**	0.591	0.388
MI-SMF [39]	0.417	0.362	0.584	0.258	0.721	0.469	0.574	0.388
MI-ACE [39]	0.414	0.361	0.531	0.254	0.723	0.477	0.570	0.384
*e*FUMI [35]	0.396	**0.384**	0.594	0.258	0.716	0.398	0.594	0.349
EM-DD [32]	0.417	0.376	0.431	0.260	0.703	0.478	0.480	0.384
mi-SVM [33]	0.265	0.207	0.247	0.201	0.328	0.224	0.248	0.331
MILES [34]	0.014	0.187	0.231	0.114	0.142	0.231	0.116	0.122

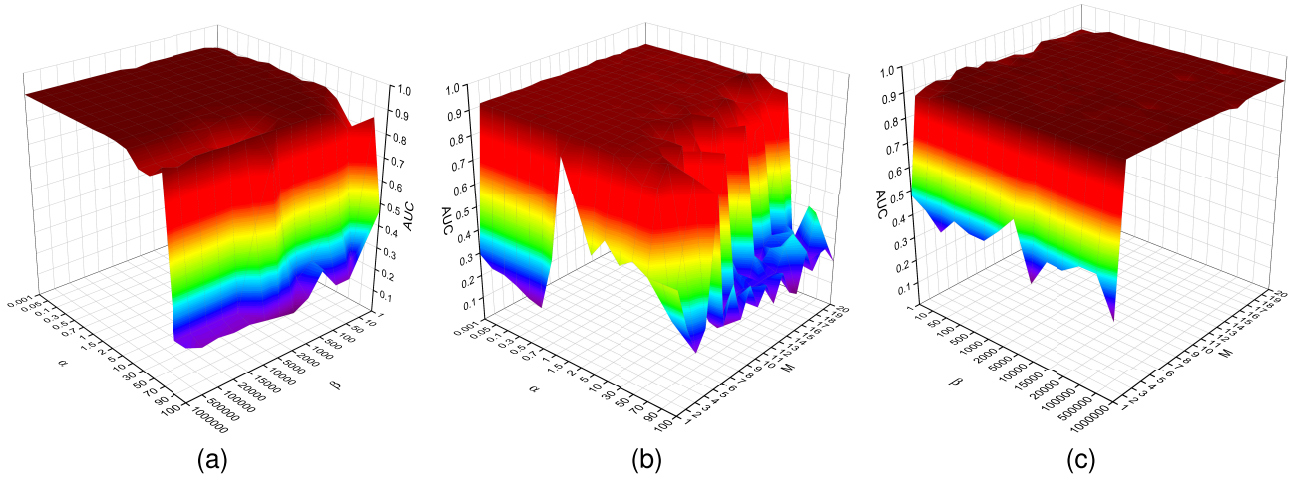


Fig. 12. Detection performance (AUC values) of MI-SPACE under (a) different combinations of α and β with a fixed $M = 3$, (b) different combinations of α and M with a fixed $\beta = 1000$, and (c) different combinations of β and M with a fixed $\alpha = 1$.

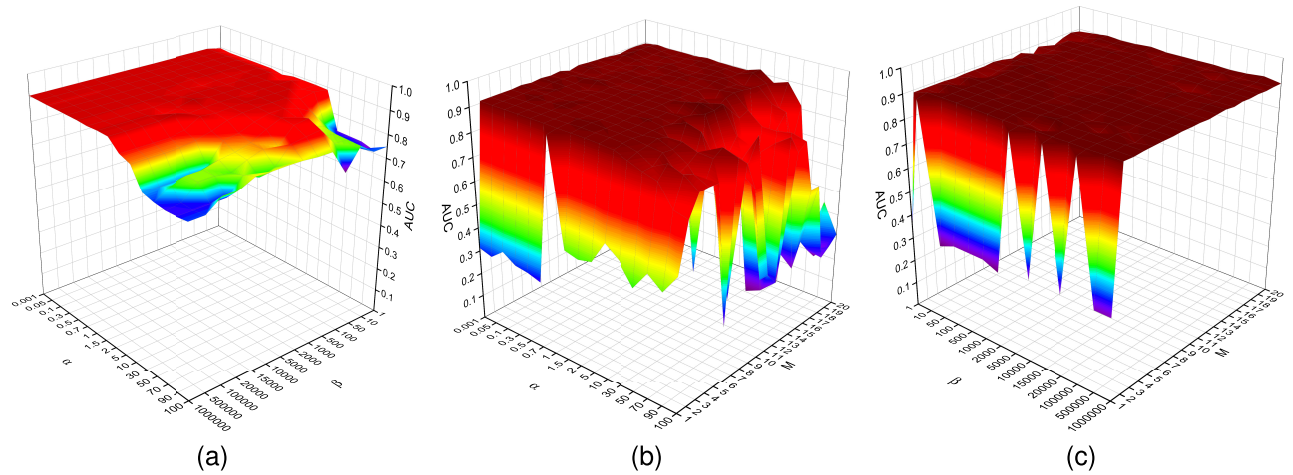


Fig. 13. Detection performance (AUC values) of MI-SPSMF under (a) different combinations of α and β with a fixed $M = 3$, (b) different combinations of α and M with a fixed $\beta = 1000$, and (c) different combinations of β and M with a fixed $\alpha = 1$.

testing. The experiment results are shown in Table XI. Generally, the proposed MI-SPACE and MI-SPSMF still achieve the overall superior performance compared with the state-of-the-art methods. We believe that generating MIL data bags with ACE detector and prior ground-truth knowledge greatly relaxes the requirements of data annotation and achieves competitive performance with using GPS information and expect that this work can inspire more investigation in MIL bag generation.

C. HyMap Hyperspectral Data

To further verify the effectiveness of the proposed methods, we included another real-field multitype subpixel target detection task that conducted on the HyMap dataset [53]. The data were collected by the HyMap sensor in July 2006 in Cook City, MT, USA. Each hyperspectral image contains 280 × 800 pixels with 126 spectral bands corresponding to wavelength 450–2500 nm at a 15-nm sampling interval. The

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TABLE XII
INFORMATION OF PLACED TARGETS IN THE HYMAP DATASET

Target ID	Target Type	Target Size	Num. of Pixels
F1	Red cotton	$3m \times 3m$	9
F2	Yellow nylon	$3m \times 3m$	9
F3a	Blue cotton	$2m \times 2m$	9
F3b	Blue cotton	$1m \times 1m$	1
F4a	Red nylon	$2m \times 2m$	9
F4b	Red nylon	$1m \times 1m$	1
V1	Chevy Blazer	N.A.	1
V2	Toyota T100	N.A.	1
V3	Subaru GL Wagon	N.A.	1

TABLE XIII
AREA UNDER THE ROC CURVES FOR MI-SPACE, MI-SPSMF, AND COMPARISON ALGORITHMS ON THE HYMAP DATASET. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	AUC	Algorithm	AUC
MI-SPSMF	**0.970**	eFUMI [35]	0.931
MI-SPACE	<u>0.967</u>	EM-DD [32]	0.934
MI-SMF [39]	0.944	mi-SVM [33]	0.906
MI-ACE [39]	0.960	MILES [34]	0.902

spatial resolution is 3 m. In the HyMap dataset, four types of fabric panels and three types of vehicles are placed as targets of interest, as listed in Table XII. Fig. 10(a) shows the false-color picture of HyMap hyperspectral scene, and Fig. 10(b) shows the ground-truth locations of each target with different colors and different sizes.

Two images over the same scene in the HyMap data, named self-test set and blind test set, were chosen for validation. The self-test set contains pixel-wise labels of targets, but the blind test set is lack of ground-truth information. Thus, in the blind test set, we manually drew a 5×5 rectangle region as an imprecisely labeled positive bag for each target by collaboratively referring to the target locations in the self-test set. The rest image area that does not have any type of target was grouped as a nontarget region. Considering that the side length (3 m) of the largest target is only approximately equal to the spatial resolution (3 m) of the image, most target instances are subpixels. Therefore, a multiclass MIL hyperspectral subpixel target detection task has been formulated for evaluating our methods. The blind test set with imprecise bag-level labels was adopted for training, and the self-test set with pixel-level ground truth was adopted for testing.

The parameter settings of MI-SPACE and MI-SPSMF for this experiment are $\alpha = 0.5$, $\beta = 1 \times 10^6$, and $M = 10$. The area under the receiver operating characteristic curve (AUC) was adopted as the performance evaluation metric. Table XIII summarizes the median AUC value over five runs. ROC curves with confidence intervals of FAR shown in Fig. 11(c) and (d) demonstrate that the proposed MI-SPACE and MI-SPSMF have a notably statistical difference compared with other approaches. From the perspective of confidence intervals of PD shown in Fig. 11(a) and (b), the proposed MI-SPACE and MI-SPSMF still achieved competitive performance. In addition, the proposed methods achieved overall higher AUC values. Thus, it can be concluded that the proposed MI-SPACE and MI-SPSMF have achieved the best overall performance on this real-field dataset, especially outperforming the comparison algorithms at a low FAR range.

TABLE XIV
AREA UNDER THE ROC CURVES FOR SPSMF, SPACE, AND COMPARISON DETECTORS ON THE HYMAP DATASET. THE BEST RESULT IS SHOWN IN BOLD AND THE SECOND BEST RESULTS ARE UNDERLINED

Algorithm	AUC	Algorithm	AUC
SPSMF+Ground Truth+VCA	<u>0.962</u>	SMF+Ground Truth [12]	0.941
SPACE+Ground Truth+VCA	**0.963**	ACE+Ground Truth [10]	0.952

For the HyMap dataset, a comparison of detectors was performed as listed in Table XIV, which also shows that the detection statistic by the proposed detectors with ground-truth spectra still outperformed the comparison algorithms.

V. ANALYSIS OF ROBUSTNESS

In this section, we first conducted a grid search experiment to analyze the robustness of the proposed MI-SPACE and MI-SPSMF with respect to different parameter settings and evaluate the impact of different terms in the proposed methods on performance. Subsequently, we investigated the robustness to different levels of SNR and subpixel mixture to demonstrate the stability of the proposed framework. Finally, we discussed the influence of spectral variability on the robustness of the proposed algorithms.

A. Analysis of Robustness to Parameter Setting

There are three parameters, α, β, and M, in the proposed methods. The parameters α and β are used to balance the contributions of the three terms in the objective function. The parameter M is the designated number of background concepts for modeling the background. To provide more profound insights into the sensitivity to parameters settings of the proposed approach, a range of different combinations of parameters were tested on the simulated data and their performance was evaluated and discussed.

The Red Slate spectrum was again specified as target endmember and the other three, Verde Antique, Phyllite, and Pyroxenite, were specified as background endmembers. The constituent endmembers and data formulation for synthetic data are listed in Table XV. For each positively labeled bag, there are only five highly mixed target points containing mean target (Red Slate) proportion $\alpha_{\text{mean}} = 0.1$ that were mixed with the other three endmembers. The number of true target pixels only takes the portion less than 1% (0.45%) of the entire positively labeled data. This setting aims to make the experiment an analogy to the real application scenario with few targets and a large amount of background instances. Gaussian white noise was added so that the SNR of the data was set to 20 dB. The varying parameters and ranges are shown in Table XVI.

Figs. 12 and 13 show the median AUC value over five runs with different parameter settings for MI-SPACE and MI-SPSMF, respectively. Since the performance property of the two algorithms with respect to parameters perturbation is

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TABLE XV
LIST OF CONSTITUENT ENDMEMBERS AND DATA FORMULATION FOR PARAMETERS PERTURBATION ANALYSIS

Bag Index	Bag Label	Instance Num.(Target Num.)	Target Endmember	Background Endmember
1-5	+	100 (5)	Red Slate	Verde Antique, Phyllite, Pyroxenite
6-10	−	1000 (0)	N/A	Verde Antique, Phyllite, Pyroxenite

TABLE XVI
TESTING RANGES FOR ANALYSIS OF MI-SPACE AND MI-SPSMF PARAMETER SETTINGS WITH ONE PARAMETER FIXED

Parameters	Testing Ranges
α	0.001, 0.05, 0.1, 0.3, 0.5, 0.7, 1, 1.5, 2, 5, 10, 30, 50, 70, 90, 100
β	1, 10, 50, 100, 500, 1000, 2000, 5000, 10000, 15000, 20000, 100000, 500000, 1000000
M	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20

TABLE XVII
TESTING RANGES FOR 3-D PARAMETER PERTURBATION ANALYSIS

Parameters	Testing Ranges
α	0.1, 0.3, 0.5, 0.7, 1, 1.5
β	500, 1000, 2000, 5000, 10000
M	3, 4, 5, 6

similar, for the sake of brevity, we only take the performance of MI-SPACE as an illustration for parameters perturbation analysis. For MI-SPACE, the detection performance under different combinations of α and β with fixed $M = 3$, α and M with fixed $\beta = 1000$, and β and M with fixed $\alpha = 1$ is shown in Fig. 12(a)–(c), respectively.

Experimental results show that our algorithms have stable detection performance in a wide parameter selection range. It can be clearly observed in Fig. 12 that the proposed methods can achieve stable and satisfactory results when α in $[0.05, 2]$, β in $\left[50, 1 \times 10^6\right]$, and $M \geq 3$. Specifically, since the true number of background endmember is 3, it is reasonable that M needs to be equal or greater than the number of real background concepts to achieve satisfactory performance. α is the scaling factor to balance the influence of target and background discrimination in the objective. Since these two terms have been normalized by the numbers of bags and/or pixels contained in each bag, $[0.05, 2]$ would be a reasonable choice according to Fig. 12(a) and (b). β is used to control the contribution of the fidelity term, which provides background modeling information for learning background concepts. Since this term is also normalized by the total number of nontarget instances and the reconstruction error of a nontarget instance could approach to zero, i.e., $\|\mathbf{x} - \mathbf{D}^{-}\mathbf{p}\| \to 0$, the setting of β prefers large values and there also exists a wide range, e.g., $[50, 1 \times 10^6]$. One can see that the detection performance is degraded and fluctuated slightly when β is too small because the insufficient influence of fidelity term cannot provide enough background modeling ability. The AUC values are 0.891 at $\beta = 0$ and 0.911 at $\beta = 5000$ with fixed $\alpha = 1$ and $M = 3$ as shown in Fig. 14, which proves the effectiveness and merits of this fidelity term.

To further evaluate the robustness of MI-SPACE and MI-SPSMF to parameters perturbation, a 3-D parameter perturbation analysis over a subset of the recommended parameter range shown in Table XVII was conducted. The proposed methods have been tested with all the possible combinations

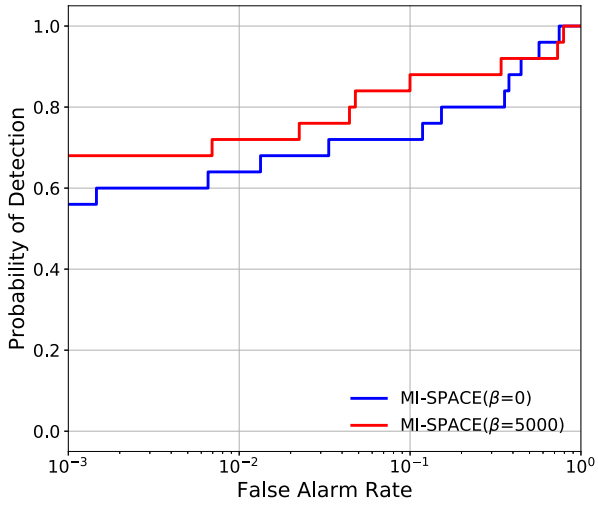


Fig. 14. ROC curves for MI-SPACE at $\beta = 0$ and $\beta = 5000$, respectively, with fixed $\alpha = 1$ and $M = 3$.

of the three parameters in Table XVII and the best and worst performance was examined. The best and the worst performance (median AUC value over five runs) are 0.912 and 0.909 for MI-SPACE and 0.910 and 0.904 for MI-SPSMF, respectively. The difference of performance with respect to this wide range of parameters is less than 1% for both methods, further verifying the robustness and stability of the proposed framework.

B. Analysis of Robustness to SNR

In this section, we conducted a series of validation experiments to investigate the robustness of detection performance respect to different levels: of 1) SNR and 2) subpixel mixture. The SNR [6] is defined as follows:

$$\text{SNR} \equiv 10\text{Log}_{10}\frac{\mathbb{E}\left[\mathbf{x}^{\mathrm{T}}\mathbf{x}\right]}{\mathbb{E}\left[\mathbf{n}^{\mathrm{T}}\mathbf{n}\right]} \tag{26}$$

where $\mathbf{x}$ and $\mathbf{n}$ represent the hyperspectral signature and additive noise vector, respectively, and $\mathbb{E}$ denotes the expectation operator.

The data formulation and constituent endmembers for this synthetic experiment are listed in Table XV, which is the same as the parameters perturbation analysis presented above.

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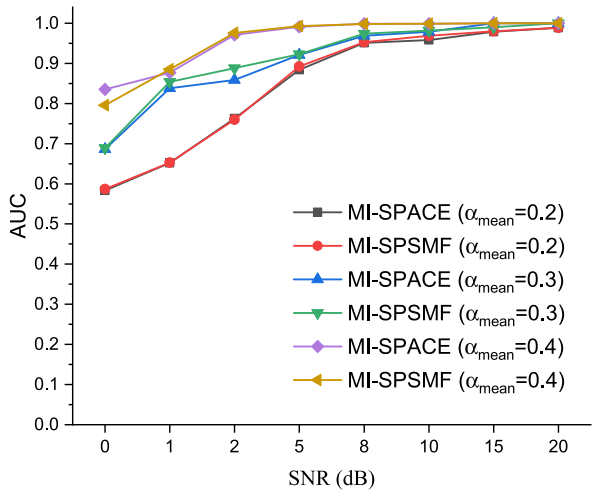


Fig. 15. Detection performance (median AUC value over five runs) as the SNR ratio increases in the varying range of {0, 1, 2, 5, 8, 10, 15, 20} dB with fixed target proportion α_{mean} (at 0.2, 0.3, and 0.4) by MI-SPACE and MI-SPSMF.

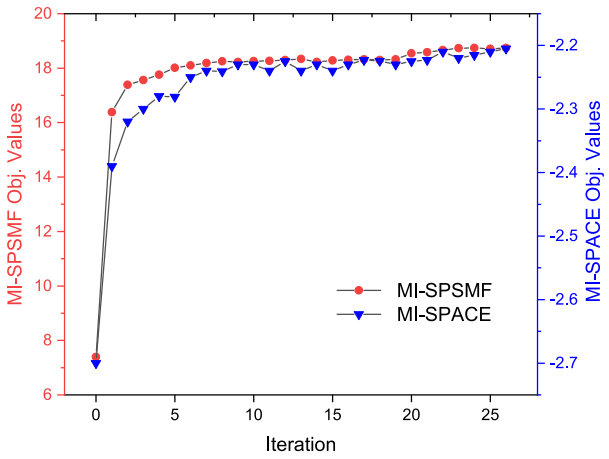


Fig. 16. Convergence curves for MI-SPSMF (red) and MI-SPACE (blue), respectively, on simulated data with target proportion $\alpha_{\text{mean}} = 0.2$ and SNR = 20 dB.

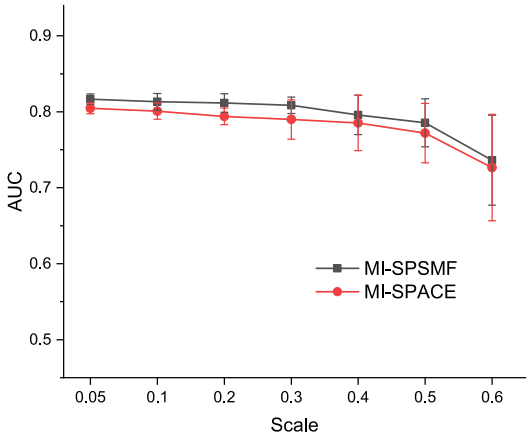


Fig. 17. Detection statistics (mean AUC values and standard deviations) of MI-SPSMF and MI-SPACE with varying degrees of endmember scaling.

Different degrees of Gaussian white noise have been added so that the simulated data with different SNR ratios have been generated. In order to clearly examine the impact of different noise levels and target proportion α_{mean} on the performance of the proposed methods, we fixed the target proportion α_{mean} to 0.2, 0.3, and 0.4 and changed the SNR in the varying range of {0, 1, 2, 5, 8, 10, 15, 20} dB. The parameters for MI-SPACE and MI-SPSMF were set to $\alpha = 0.5$, $\beta = 1 \times 10^3$, and $M = 3$.

Fig. 15 shows the trend of detection performance (median AUC value over five runs) as the SNR increases with fixed target proportion α_{mean} (at 0.2, 0.3, and 0.4) provided by MI-SPACE and MI-SPSMF. It can be found in Fig. 15 that the detection performance of both methods with respect to different levels of mean target proportion increases with the increase of SNR and approaches to a stable range when the SNR exceeds a certain threshold, e.g., 5 dB. The conclusion can be made that the performance of the proposed methods on different levels of target presence is not severely affected by noise if the data quality is higher than a bottom line.

The convergence curves for MI-SPSMF and MI-SPACE on simulated data with target proportion $\alpha_{\text{mean}} = 0.2$ and SNR = 20 dB were plotted in Fig. 16. The convergence curves show an upward trend since the objectives of the proposed methods are to maximize the detection statistics of positive bags and minimize the detection probabilities and reconstruction errors of negative bags. It can also be found that the curves increase rapidly first and then stabilize as the number of iterations increases, which provides solid evidence to the convergence and effectiveness of the proposed methods.

C. Analysis of Robustness to Spectral Variability

The effectiveness and performance of the learned target signature cannot be guaranteed due to the inherent variability of endmembers [6], [54]–[56]. To investigate the robustness of the detection performance for the proposed methods to spectral variability, the synthetic dataset with varied degrees of spectral variability has been designed and performance on these datasets has also been evaluated and discussed.

Similarly, the Red Slate spectrum was specified as target endmember and the other three, Verde Antique, Phyllite, and Pyroxenite, were specified as background endmembers. The constituent endmembers and data formulation for synthetic data are listed in Table XV (same as parameters perturbation experiments). The mean target proportion α_{mean} is set to 0.1. Each simulated pixel with spectral variability has been simulated according to endmember-dependent scaling factors and complex noise [56]. For each simulated spectrum, different from the previous simulated data generation process, we first multiplied each original endmember by a randomly selected endmember-dependent scaling factor ranging in [1 − scale, 1 + scale], and then, the Gaussian white noise was added so that the SNR of each scaled endmember is 15 dB. Afterward, we mixed those scaled noisy endmembers following the linear mixing model [15] with abundances sampled from the Dirichlet distribution. A 15-dB Gaussian white noise was added to the mixed spectra again.

During this experiment, we changed scale in the varying range of {0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6}. For each scale, five training datasets were generated separately using the same

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generating procedure. The estimated target signatures were tested on one dataset generated by four original endmembers with 0.1 mean target proportion. Fig. 17 shows the detection performance (mean AUC values and standard deviations) of MI-SPACE and MI-SPSMF. In general, it can be found in Fig. 17 that the performance of the proposed methods is not severely affected by spectral variability. Both of the proposed approaches achieve stable detection performance when the scale factor is lower than 0.4. Considering that scale = 0.4 (the original endmembers are scaled by random factors in the range of [0.6, 1.4], along with complex noises) can bring significant spectral deviation to the simulated data, one can conclude that the proposed algorithms are robust to moderate potential spectral variabilities.

VI. Conclusion

Two subpixel discriminative target characterization methods, MI-SPACE and MI-SPSMF, as well as the corresponding subpixel detectors SPACE and SPSMF, have been proposed and investigated in this article for addressing the task of subpixel target detection with imprecise training labels. In particular, the proposed methods integrated the concept of background modeling to discriminative subpixel target characterization and achieved superior performance on both synthetic and real-field challenging hyperspectral target detection tasks.

Appendix A
Derivation of SPACE and SPSMF Subpixel Detectors

This section provides a detailed derivation of the proposed SPACE and SPSMF detectors from spectral space to whitened space. Specifically, the proposed SPACE detector can be written as follows:

$$R_{\text{SPACE}}(\mathbf{s}, \mathbf{r}_s) = \frac{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \mathbf{r}_s}{\sqrt{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \mathbf{s}} \sqrt{\mathbf{r}_s^T \boldsymbol{\Sigma}_b^{-1} \mathbf{r}_s}} = \frac{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \left(\mathbf{x} - \mathbf{D}^- \mathbf{a}^-\right)}{\sqrt{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \mathbf{s}} \sqrt{(\mathbf{x} - \mathbf{D}^- \mathbf{a}^-)^T \boldsymbol{\Sigma}_b^{-1} (\mathbf{x} - \mathbf{D}^- \mathbf{a}^-)}} \tag{A.1}$$

where $\mathbf{r}_s = \mathbf{x} - \mathbf{D}^- \mathbf{a}^-$ is the residual target component. Subsequently, $\boldsymbol{\Sigma}_b^{-1}$ could be decomposed by solving the eigenvalue decomposition of $\boldsymbol{\Sigma}_b$

$$\begin{aligned} \boldsymbol{\Sigma}_b^{-1} &= \mathbf{U} \mathbf{P}^{-1} \mathbf{U}^T \\ &= \mathbf{U} \mathbf{P}^{-\frac{1}{2}} \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \end{aligned} \tag{A.2}$$

where $\mathbf{U}$ and $\mathbf{P}$ are the eigenvectors and eigenvalues of $\boldsymbol{\Sigma}_b$.

In consideration of that $\mathbf{r}_s$ is a function of $\mathbf{x}$ and $\mathbf{D}^-$, we adopt $R_{\text{SPACE}}([\mathbf{s}\ \mathbf{D}^-], \mathbf{x})$ to replace $R_{\text{SPACE}}(\mathbf{s}, \mathbf{r}_s)$ in the following derivation. Based on (A.2), SPACE can be rewritten as

$$\begin{aligned} R_{\text{SPACE}}([\mathbf{s}\ \mathbf{D}^-], \mathbf{x}) &= \frac{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \mathbf{r}_s}{\sqrt{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \mathbf{s}} \sqrt{\mathbf{r}_s^T \boldsymbol{\Sigma}_b^{-1} \mathbf{r}_s}} \\ &= \frac{\mathbf{s}^T \mathbf{U} \mathbf{P}^{-\frac{1}{2}} \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{r}_s}{\sqrt{\mathbf{s}^T \mathbf{U} \mathbf{P}^{-\frac{1}{2}} \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{s}} \sqrt{\mathbf{r}_s^T \mathbf{U} \mathbf{P}^{-\frac{1}{2}} \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{r}_s}} \\ &= \left(\frac{\hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|}\right)^T \left(\frac{\hat{\mathbf{r}}_s}{\|\hat{\mathbf{r}}_s\|}\right) \\ &= \hat{\hat{\mathbf{s}}}^T \hat{\hat{\mathbf{r}}}_s \end{aligned} \tag{A.3}$$

where

$$\hat{\hat{\mathbf{s}}} = \frac{\hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|}, \quad \hat{\mathbf{s}} = \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{s} \tag{A.4}$$

$$\hat{\hat{\mathbf{r}}}_s = \frac{\hat{\mathbf{r}}_s}{\|\hat{\mathbf{r}}_s\|}, \quad \hat{\mathbf{r}}_s = \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{r}_s. \tag{A.5}$$

In addition, the difference between SPACE and SPSMF is that SPSMF does not normalize the whitened target component $\hat{\mathbf{r}}_s$. Thus, SPSMF can be rewritten in a similar way as follows:

$$\begin{aligned} R_{\text{SPSMF}}([\mathbf{s}\ \mathbf{D}^-], \mathbf{x}) &= \frac{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \mathbf{r}_s}{\sqrt{\mathbf{s}^T \boldsymbol{\Sigma}_b^{-1} \mathbf{s}}} \\ &= \frac{\mathbf{s}^T \mathbf{U} \mathbf{P}^{-\frac{1}{2}} \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{r}_s}{\sqrt{\mathbf{s}^T \mathbf{U} \mathbf{P}^{-\frac{1}{2}} \mathbf{P}^{-\frac{1}{2}} \mathbf{U}^T \mathbf{s}}} \\ &= \left(\frac{\hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|}\right)^T \hat{\mathbf{r}}_s \\ &= \hat{\hat{\mathbf{s}}}^T \hat{\mathbf{r}}_s. \end{aligned} \tag{A.6}$$

Appendix B
Derivation of Target Characterization Updates

The joint MIL objective J for the proposed subpixel target characterization method is shown in (A.7)

$$\begin{aligned} J &= J_{\text{det}}([\mathbf{s}\ \mathbf{D}^-], \mathbf{B}) - \beta J_{\text{fid}}(\mathbf{D}^-, \mathbf{B}_i | L_i = 0) \\ \text{s.t. } & \hat{\hat{\mathbf{s}}}^T \hat{\hat{\mathbf{s}}} = 1. \end{aligned} \tag{A.7}$$

Based on (A.7), the optimization of MI-SPACE and MI-SPSMF can be decomposed into two subproblems, updating the concepts set $\mathbf{D}$ and the convex unmixing of the input data using the FCLS unmixing method given $\mathbf{D}$ or $\mathbf{D}^-$. The optimization problem of $\mathbf{D}$ could be decomposed into target signature learning for $\mathbf{s}$ and background concepts learning for $\mathbf{D}^-$.

A. Optimization Process of Target Signature $\mathbf{s}$

The update equation for target signature $\mathbf{s}$ can be solved by applying the associated Lagrangian

$$\begin{aligned} \mathcal{L} &= J_{\text{det}}([\mathbf{s}\ \mathbf{D}^-], \quad \mathbf{B}) - \beta J_{\text{fid}}(\mathbf{D}^-, \mathbf{B}_i | L_i = 0) \\ &\quad - \lambda\left(\hat{\hat{\mathbf{s}}}^T \hat{\hat{\mathbf{s}}} - 1\right) \\ &= \frac{1}{K^+} \sum_{i=1}^{K^+} R([\mathbf{s}, \mathbf{D}^-], \mathbf{x}_i^*) \\ &\quad - \frac{\alpha}{K^-} \sum_{i=K^++1}^{K} \frac{1}{N_i} \sum_{j=1}^{N_i} R([\mathbf{s}, \mathbf{D}^-], \mathbf{x}_{ij}) \\ &\quad - \frac{\beta}{2N^-} \sum_{i=K^++1}^{K} \sum_{j=1}^{N_i} ||\mathbf{x}_{ij} - \mathbf{D}^- \mathbf{p}_{ij}||^2 - \lambda\left(\hat{\hat{\mathbf{s}}}^T \hat{\hat{\mathbf{s}}} - 1\right) \end{aligned} \tag{A.8}$$

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$$\begin{aligned}\frac{\partial J_{\text{SPACE}}}{\partial \mathbf{d}_m} &= \frac{\partial \hat{\mathbf{d}}_m}{\partial \mathbf{d}_m}\frac{\partial\left(J_{\text{det}}([\mathbf{s}\ \mathbf{D}^-],\mathbf{B})\right)}{\partial \hat{\mathbf{d}}_m} - \frac{\partial\left(\beta J_{\text{fid}}(\mathbf{D}^-,\ \ \mathbf{B}_i|L_i=0)\right)}{\partial \mathbf{d}_m} \\ &= \mathbf{U}\mathbf{P}^{-\frac{1}{2}}\frac{\partial}{\partial \hat{\mathbf{d}}_m}\left(\frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{s}}^T\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\mathbf{s}}^T\hat{\mathbf{r}}_{sij}\right) - \frac{\partial}{\partial \mathbf{d}_m}\left(\frac{\beta}{2N^-}\sum_{i=K^++1}^{K}\sum_{j=1}^{N_i}||\mathbf{x}_{ij}-\mathbf{D}^-\mathbf{p}_{ij}||^2\right) \\ &= \mathbf{U}\mathbf{P}^{-\frac{1}{2}}\left[\frac{1}{K^+}\sum_{i=1}^{K^+}\frac{a_{im}^*}{\left\|\hat{\mathbf{r}}_{si}^*\right\|}\left(\hat{\mathbf{s}}^T\hat{\mathbf{r}}_{si}^*\hat{\mathbf{r}}_{si}^* - \hat{\mathbf{s}}\right) + \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\frac{a_{ijm}}{\left\|\hat{\mathbf{r}}_{sij}\right\|}\left(\hat{\mathbf{s}} - \hat{\mathbf{s}}^T\hat{\mathbf{r}}_{sij}\hat{\mathbf{r}}_{sij}\right)\right] \\ &\quad + \frac{\beta}{N^-}\sum_{i=K^++1}^{K}\sum_{j=1}^{N_i}p_{ijm}(\mathbf{x}_{ij}-\mathbf{D}^-\mathbf{p}_{ij})\end{aligned} \tag{A.17}$$

where λ is the Lagrange multiplier. Specifically, by substituting the SPACE detection response R_{SPACE} shown in (A.3), one can get the Lagrangian $\mathcal{L}_{\text{SPACE}}$ with respect to MI-SPACE

$$\begin{aligned}\mathcal{L}_{\text{SPACE}} = &\frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{s}}^T\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\mathbf{s}}^T\hat{\mathbf{r}}_{sij} \\ &- \frac{\beta}{2N^-}\sum_{i=K^++1}^{K}\sum_{j=1}^{N_i}||\mathbf{x}_{ij}-\mathbf{D}^-\mathbf{p}_{ij}||^2 - \lambda\left(\hat{\mathbf{s}}^T\hat{\mathbf{s}}-1\right).\end{aligned} \tag{A.9}$$

The gradient of the Lagrangian $\mathcal{L}_{\text{SPACE}}$ with respect to $\hat{\mathbf{s}}$ can be computed as

$$\frac{\partial \mathcal{L}_{\text{SPACE}}}{\partial \hat{\mathbf{s}}} = \frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\mathbf{r}}_{sij} - 2\lambda\hat{\mathbf{s}}. \tag{A.10}$$

Set (A.10) to zero

$$\frac{\partial \mathcal{L}_{\text{SPACE}}}{\partial \hat{\mathbf{s}}} = 0. \tag{A.11}$$

Then, the analytical solution of $\hat{\mathbf{s}}$ can be derived as

$$\hat{\mathbf{s}} = \frac{1}{2\lambda}\left(\frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\mathbf{r}}_{sij}\right). \tag{A.12}$$

Define $\mathbf{t}$ as

$$\mathbf{t} = \frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\mathbf{r}}_{sij}. \tag{A.13}$$

Subsequently, the value of λ can be determined by enforcing the constraint $\hat{\mathbf{s}}^T\hat{\mathbf{s}} = 1$

$$\lambda = \frac{\|\mathbf{t}\|}{2}. \tag{A.14}$$

The update equation of $\hat{\mathbf{s}}$ with respect to MI-SPACE can be formulated as (A.15)

$$\hat{\mathbf{s}} = \frac{\mathbf{t}}{\|\mathbf{t}\|} \text{ where } \mathbf{t} = \frac{1}{K^+}\sum_{i=1}^{K^+}\hat{\mathbf{r}}_{si}^* - \frac{\alpha}{K^-}\sum_{i=K^++1}^{K}\frac{1}{N_i}\sum_{j=1}^{N_i}\hat{\mathbf{r}}_{sij}. \tag{A.15}$$

For MI-SPSMF, the update equation of $\hat{\mathbf{s}}$ can be obtained in a similar manner shown in (22).

B. Optimization Process of Background Concepts Set $\mathbf{D}^-$

The optimization of background concepts for MI-SPACE or MI-SPSMF is performed by taking gradient ascent with respect to one background concept at a time and holding the rest concepts fixed. The gradient of $\mathbf{d}_m$ is computed in the following:

$$\frac{\partial J}{\partial \mathbf{d}_m} = \frac{\partial\left(J_{\text{det}}([\mathbf{s}\ \mathbf{D}^-],\mathbf{B})\right)}{\partial \mathbf{d}_m} - \frac{\partial\left(\beta J_{\text{fid}}(\mathbf{D}^-,\mathbf{B}_i|L_i=0)\right)}{\partial \mathbf{d}_m}. \tag{A.16}$$

For MI-SPACE, the gradient of the MIL objective function J_{SPACE} with respect to a background concept $\mathbf{d}_m$ can be further derived as (A.17), as shown at the top of the page. The gradient $(\partial J_{\text{SPSMF}}/\partial \mathbf{d}_m)$ for MI-SPSMF can be obtained in a similar way shown as (24).

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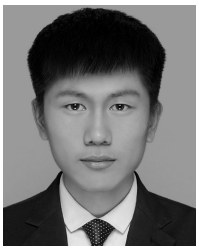
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